

Rotational Reduction of Quasi-Spherical Sulfuryl Fluoride (SO₂F₂): Gauge Freedom, Quartic Indeterminacy, and Truncation Artifacts

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Abstract

Sulfuryl fluoride, SO₂F₂, is a quasi-spherical asymmetric top with rotational constants $A \approx B \approx C$ and an exceptionally small sphericity parameter, $\gamma = 2(C - B)/(C + B) \approx -0.003$. Within Watson's theory, only five independent combinations of the six quartic centrifugal-distortion coefficients are spectroscopically determinable. However, the conventional A - and S -reductions, defined by unitary transformation $\hat{U} = \exp(i\hat{S}_3(s_{111}))$, become poorly conditioned. Sarka *et al.* (2000) argued that, near the spherical-top limit, the small rotational denominators invalidate the conventional A - and S -reduction constraints and leave the six quartic coefficients only marginally affected by an admissibly small contact transformation; they therefore concluded that all six quartic (and nine sextic) centrifugal-distortion constants are spectroscopically determinable. We re-evaluate this statement and show that the contact transformation generates a one-dimensional gauge orbit along which the rotational constants and quartic coefficients change, while the spectrum remains invariant through the first order. We further show that the proposed condition $P'_{006} \approx 0$ to determine s_{111} is not justified because the omitted contributions are required for the exact cancellation. The observed narrow uncertainties are conditional on the sextic constraint $h_1 = 0$. Furthermore, we derive a first-order invariant $R_5 + 2\sigma R_6 = C_{R_5 R_6}$, where $\sigma = (2B_z - B_x - B_y)/(B_x - B_y)$. Theoretical conclusions are supported by numerical tests, demonstrating consistent change in six quartic constants with alteration of the asymmetry parameter κ . Finally, the ratio σ provides a first-order conditioning criterion: $|\sigma| \ll 1$ favors the A -reduction, whereas $|\sigma| \gg 1$ favors the S -reduction. For SO₂F₂ in the I^r representation, $\sigma = 8.7$ selects the S -reduction.

Keywords: centrifugal distortion, Watson reduction, contact transformation, gauge freedom, quasi-spherical top

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1. Introduction

Sometimes, an application of well-established and widely used theories of nuclear motion in molecules to new objects produces unexpected surprises and poses puzzles to theoreticians. For example, Watson's theory of reduced effective rotational Hamiltonians [1–4] at first only considered general asymmetric top molecules without accidentally close rotational constants ($A \neq B \neq C$), for which the so-called A -reduction was developed. An essential modification (S -reduction) for a quasi-symmetric top ($A \approx B$ or $B \approx C$) was developed later by Winnewisser *et al.* (1972) [5] and following works [6, 7]. Nevertheless, even more usual situation of an accidentally spherical (quasi-spherical) limit with $A \approx B \approx C$ may occur. This is the case for the sulfonyl fluoride molecule, SO_2F_2 , which is an asymmetric top with C_{2v} point symmetry. Indeed, as it was found by Lide *et al.* (1957) in a MW study [8], the ground state rotational constants of $^{32}\text{S}^{16}\text{O}_2\text{F}_2$ are $A = 5134.26 \pm 0.5$, $B = 5073.04 \pm 0.01$, $C = 5057.04 \pm 0.01$ MHz, so that these constants nearly coincide, with a sphericity parameter being $\gamma = 2(C - B)/(C + B) \approx -0.003$ [9]. In this situation both types of reductions are formally not suitable because the condition of smallness of the first rotational unitary transformation $|s_{111}| \lesssim T_{\alpha\beta}^*/B_\alpha^*$ [10–12] (asterisks denote typical values of these constants) is not fulfilled and the perturbation theory is therefore poorly conditioned.

In 2000, Sarka, Demaison, Margulès *et al.* [13] published a research paper entitled ‘Centrifugal Distortion Analysis of a Near-Spherical Top, SO_2F_2 : The First Determination of All Six Quartic Centrifugal Distortion Constants for an Asymmetric Top’. In this work, the authors assembled a system of arguments in supporting the statement, claimed in the Abstract: ‘*we wanted to verify our theoretical prediction that for such a molecule all six quartic centrifugal distortion constants (and nine sextic distortion constants) should be determinable, while for a standard asymmetric rotor, Watson has shown that only five quartic and seven sextic distortion constants are determinable. The analysis of the spectra confirmed our predictions, because all six quartic constants were well determinable.*’ This approach, called 6-reduction, may be interpreted as if the full set of quartic constants is non-redundant and corresponds to true physically meaningful constants. In support of this construction, Sarka *et al.* provided a number of arguments, including the numerical evidence of narrow confidence intervals of fitted parameters, as well as demonstrated good correspondence of fitted parameters with theoretical *ab initio* unreduced six quartic constants.

Several subsequent publications on high-resolution spectra of SO_2F_2 did not object the main statement of Sarka *et al.* of a possibility to determine all six quartic constants [9, 12, 14–21], although various alternative treatments to quasi-sphericity of SO_2F_2 were proposed, mainly using the theory of irreducible tensor operators (ITO). Nevertheless, the physical and mathematical limitations of Sarka *et al.*'s interpretation have remained largely unaddressed.

In this article, we have rigorously demonstrated that due to existence of a $\hat{S}_3(s_{111})$ -controlled unitary equivalence of the rotational Hamiltonian family (known in physics as the *gauge freedom*), an asymmetric top, possessing six quartic centrifugal distortion operators, has one of them physically indistinguishable from a momentum-dependent reorientation of the effective molecule-fixed frame in the semiclassical picture, which reduces the number of experimentally determinable parameters to five only. This fundamental property of a non-rigid rotating asymmetric top does not disappear when a formal perturbation theory treatment becomes poor-conditioned. We consistently analyze both Sarka *et al.*'s argumentation and briefly review the underlying physical theory that is scattered in foundational works and is seemingly insufficiently well covered in essential Watson's works and other monographs [1–4, 22–24].

Sarka *et al.* makes a central conclusion that, since both A and S -reductions produce abnormally large values of the first-order perturbation theory parameters s_{111} , none of six quartic constants can be fixed to zero and consequently all of them are experimentally determinable. Some solutions, proposed by Sarka *et al.*, are non-standard and require a specific consideration. In particular, it was

suggested to define the first-order parameter s_{111} of unitary transformations in the second order from a simplified expression for the unitary transformed sextic parameter Φ'_{006} . Other observed artifacts, e.g. narrow confidence intervals $\sigma(H'_k)$ for effective Hamiltonians with linear dependency, are falling to rather numerical methods theory and were not ever discussed by Watson.

At the beginning of Sarka *et al.* paper it is fully acknowledged that, according to Watson's theory, among the six quartic distortion constants $T_{klm}[(\mathbf{J}^2)^{k/2}, J_z^l, (J_+^m + J_-^m)]$, only their five combinations are determinable from the spectra, and the indeterminacies are caused by existence of the diagonal rotational contact transformation $H' = \exp(i\hat{S}) \cdot H \cdot \exp(-i\hat{S})$ that preserves eigenvalues of the rotational Hamiltonian. However, the subsequent part of this works concludes and provides numerous arguments in favor of the statement that when a molecule is closely approaching a quasi-spherical limit, the number of experimentally determinable parameters may become bigger. In this paper, we conduct a deep theoretical analysis of Sarka's *et al.* argumentation and draw a central conclusion that the numerical artifacts, obtained in that paper were given an incorrect interpretation.

After presenting a brief review of Watson's reduction in the first and second order Sarka *et al.* stated that 'the contributions of the unitary transformation to the quartic distortion constants are small and these constants are only marginally affected by the transformation'. Besides, small values of denominators $\Delta_{xy} = (B_x - B_y)$ and $\Lambda_{xyz} = (2B_z - B_x - B_y)$ for both types of reductions (A, S) and consequently abnormally large values of parameters s_{111} were interpreted as if none of the quartic distortion constants can be constrained to the zero value, because otherwise the essential condition of convergence $s_{klm} \approx \chi^{2(k+l+m-1)}$ is not fulfilled. The central conclusion was made that 'for quasi-spherical asymmetric top, all six quartic distortion constants should be determinable from the spectra.'

Next, Sarka *et al.* stated that 'the parameter s_{111} cannot by fixed be constraining one of the quartic distortion constants to zero value' and proposed a solution that s_{111} must be found from sextic distortion constants using $\Phi'_{006} = 0$ or $\Phi'_{024} = 0$ (the latter constant was given with a typo), where only linear term in s_{111} is retained. In support of this proposition, Sarka *et al.* argued that for a quasi-spherical top with $B_x \approx B_y \approx B_z$, the contributions of higher order transformation parameters $s_{311}, s_{131}, s_{113}$, to the sextic distortion constants are very small because they always contain either Δ_{xy} or Λ_{xyz} factor and their effect on the sextic constants can be safely neglected. This overall solution, where all six quartic distortion constants are fitted and the most important reduction parameter s_{111} is fixed by a constraint in the sextic Hamiltonian, was called '6-Reduction'.

Sarka suggested to define s_{111} from the reduced form of Φ_{006} , which is a meaningless operation, as such value will not equate to zero any quartic or sextic constant. Besides, Sarka *et al.* claimed that 'when the sextic constants only depend on s_{111} , it is possible to constrain to zero value only one sextic parameter and therefore with sufficient data in principle nine sextic distortion constants can be determined for quasi-spherical asymmetric top from the spectra.'

In the experimental part of Sarka *et al.*'s work, 'the results of fits with conventional A and S reductions confirmed the theoretical prediction that these reductions are not suitable' and the 6-Reduction was finally applied. It was found that the standard deviation of the resulting fit was as good as for the S Reduction, and for six quartic and all nine sextic constants but one (h_1) the standard deviations were quite small. The constraint $h_1 = 0$ did not significantly worsen the RMSD. There is a logical contradiction in setting $h_1 = 0$ as they claim that all nine sextic centrifugal distortion constants can be determined. These narrow confidence intervals were interpreted as a proof of this Proposition.

The final Sarka *et al.*'s argument arose after comparing fitted constants with the theoretical values, obtained from the force field using analytic formulas [23]. It was found that the derived centrifugal distortion constants have correct signs and reproduce the fitted values within 15%.

98 The present article is aiming at explaining the controversy between Sarka *et al.*'s propositions
99 and the conventional theory of rotational reduction, developed by Watson. Watson's approach
100 is based on formal application of the rotational perturbation theory and the hypervirial theorem
101 [1]; he never considered the underlying fundamental physics of a rotating flexible molecule and
102 principal indeterminacy, that appears due to the so-called '*gauge freedom*' of choosing a molecule-
103 fixed frame when a molecule is distorted by centrifugal forces. It is gauge freedom that permits
104 five quartic, seven sextic, nine octic, etc. constants being rigorously determinable at correspond-
105 ing orders of perturbation theory, while a practical application of the perturbation theory may
106 be subject to various numerical situations, that may create illusions of a failure of underlying
107 physics. Here, we will briefly mention principal objections against the system of Sarka *et al.*'s
108 argumentation and in the main part of this work we will present a more systematic proofs of
109 physical considerations and mathematical facts that refute certain statements of Ref. [13].

110 First, Watson's unitary transformation $\hat{H}' = \hat{H} + i[\hat{S}_3, \hat{H}_0]$, where $\hat{S}_3 = s_{111}(\hat{J}_x\hat{J}_y\hat{J}_z + \hat{J}_z\hat{J}_y\hat{J}_x)$
111 can be considered as a generator of an infinitesimal 'wobbling' (tilting) of the coordinate frame. It
112 reflects the fact that some manifestations of centrifugal distortion are physically indistinguishable
113 from a small J -dependent rotation of the coordinate frame used to build the theoretical model.
114 This indeterminacy of the coordinate frame, called *gauge freedom*, ensures that among six quartic
115 centrifugal distortion parameters there is always one strict linear dependence and by choosing
116 a specific value of s_{111} it is possible to isolate one constant and fit five other truly physically
117 meaningful constants.

118 Second, the uncertainties σ are derived from the diagonal elements of the Covariance Matrix
119 $\mathbb{C} = (\mathbb{J}^T W \mathbb{J})^{-1}$, which is the inverse of the approximated Hessian matrix. Introducing an artificial
120 constraint to the sextic constant h_1 artificially breaks the singularity, while the least squares fitting
121 procedure (LSQ) still compensates the truncation error with all six quartic and the remaining
122 sextic constants.

123 Lastly, an *ab initio* calculation of the harmonic and anharmonic force fields in a specific
124 molecule-fixed geometric frame (usually the PAS) evaluates unreduced centrifugal distortion con-
125 stants without applying any rotational contact transformations. Sarka *et al.* used an *ab initio*
126 force field calculation as an independent test of the six quartic constants obtained from their
127 6-reduction fit [13] and the reported absolute relative discrepancies were within 15%. Together
128 with the finite formal uncertainties of all six fitted constants, the normal quartic magnitude of
129 R_5 , and its substantial contribution to the measured transition frequencies, this agreement was
130 interpreted as an independent confirmation that all six quartic constants were physically valid and
131 spectroscopically determinable.

132 This comparison, however, establishes agreement between two particular representatives of
133 the same contact transformation equivalence class, rather than the existence of six independent
134 quartic observables. Let the complete parameter vector, including the rotational and distortion
135 constants, vary along the contact transformation orbit according to

$$\mathbf{p}(s_{111}) = \mathbf{p}(0) + s_{111}\mathbf{g} + \mathcal{O}(s_{111}^2), \quad \frac{d\boldsymbol{\nu}}{ds_{111}} = \mathbb{J}\mathbf{g} \simeq 0, \quad (1)$$

136 where $\mathbf{g}$ is the gauge vector and $\mathbb{J}$ is the spectroscopic Jacobian. An *ab initio* force-field calculation
137 supplies one representative $\mathbf{p}_{\text{ai}}$ in a specified molecule-fixed convention, for which $s_{111} = 0$ may
138 be adopted by definition, whereas the condition imposed on a sextic coefficient selects another
139 representative $\mathbf{p}_{\text{fit}}$ on the same orbit. Consequently,

$$\mathbf{T}_{\text{fit}} \simeq \mathbf{T}_{\text{ai}} \quad \not\approx \quad \mathbb{J}\mathbf{g} \neq 0. \quad (2)$$

140 The observed agreement may therefore validate the selected gauge-fixed parameter set and indicate

that it lies close to the *ab initio* convention, but it cannot remove the null vector or prove that the sixth quartic coefficient is an independent spectroscopic observable.

In the practical Section ‘Calculations’ we repeat a convincing numerical experiment, developed by Dreizler and Dendl (1965) [25] for studying other subjects, to our molecule SO_2F_2 . Their calculation tested different rigid-rotor molecular geometries to fitting the observed MW data, where missing contributions were modeled by Kivelson and Wilson [26] set of $A_1 \dots A_6$ quartic centrifugal distortion parameters. We will show that for the physically meaningful molecular geometry and the corresponding varied asymmetry parameter κ_0 the contributions of centrifugal distortion energy are minimal, while outside κ_0 the functional behavior of $A_k(\kappa)$ demonstrates a strictly coordinated character, proving the existence of a hidden linear dependency. At the neighborhood of κ_0 the residual fitting error $\sigma(\kappa)$ reveals a very shallow minimum, also proving an existence of multiple solutions, equally reproducing the experimental data.

In this work, we have found, seemingly for the first time, the existence of the first-order quartic invariant combination $R_5 + 2\sigma R_6 = C_{R_5 R_6}$, where $\sigma = (2B_z - B_x - B_y)/(B_x - B_y)$. Its validity was checked using the GS MW transitions dataset from Ref. [17]. Our numerical test unequivocally proves the validity of $C_{R_5 R_6} = \text{const}$ in rather wide range for SO_2F_2 , despite of poor conditioning of perturbation theory. The found invariant has potentially a wide scope of applications and may motivate for searching of higher-order invariants, which is not accomplished in this work.

A possible reduction solution for the SO_2F_2 molecule can be fixing constants R_5 or R_6 at *ab initio* values, using PES, obtained at high-level quantum chemistry calculations, and fitting the remaining five quartic parameters. The same solution is applicable to sextic constants, where three $\Delta K = \pm 4, \pm 6$ can also be frozen at *ab initio* values.

2. Theory

The theory of reducing effective rotational Hamiltonians was developed in 1960s’-1970s’ by Watson in a fundamental series of works [1–4, 23, 27], likely inspired by an earlier work of Dreizler and Dendl (1965) [25], which provided an evidence of the existing linear dependence among centrifugal distortion parameters $A_1 \dots A_6$, appearing in the Kivelson and Wilson (1952) [26] expression $W_r(W_0, A_1 \dots A_6)$ for a rigid rotor energy, perturbed in the first order by six quartic terms. It should be noted that W_r was suspected to have linear dependencies even prior to work [25]: Hill and Edwards (1964) [28] admitted an existence of a linear dependence between a related quartic representation via Nielsen’s constants $D_J, D_{JK}, D_K, \delta_J, R_5, R_6$ [29]. Soon afterwards Watson explained [1, 3] this effect and proposed the solution by unitary transforming (*A*-reduction) the rotational Hamiltonian [2]. Since then, using of rotational Hamiltonians became a cornerstone of high-resolution rotational and vibration-rotation spectroscopy.

2.1. Sextic Rotational Hamiltonian and Its Unitary Transformations

Let Cartesian molecule-fixed components of angular momentum be satisfying the following usual commutation relation:

$$[\hat{J}_\alpha, \hat{J}_\beta] = -i\mathbf{e}_{\alpha\beta\gamma}\hat{J}_\gamma, \quad (3)$$

where $(\alpha, \beta, \gamma = x, y, z)$ and $\mathbf{e}_{\alpha\beta\gamma}$ is the Levi-Chivita symbol. The corresponding ladder operators are defined by convention as follows:

$$\hat{J}_\pm = \hat{J}_x \mp i\hat{J}_y. \quad (4)$$

For simplifying the notation, we will be denoting operators (e.g. $\hat{J}, \hat{H}, \hat{S}$) both with or without hats, as it is easy to distinguish them from scalars from a context.

182 Depending on a particular task, it is convenient to use either Cartesian form of a sextic rota-
 183 tional Hamiltonian in Watson form [23], where the notation $T_{\alpha\beta} = \frac{1}{4}\tau'_{\alpha\alpha\beta\beta}$ simplifies the expression:

$$\begin{aligned}
 H_{rot}^{(xyz)} &= H_2 + H_4 + H_6, \\
 H_2 &= \sum_{\alpha} B_{\alpha} J_{\alpha}^2 = B_x J_x^2 + B_y J_y^2 + B_z J_z^2, \\
 H_4 &= \sum_{\alpha} T_{\alpha} J_{\alpha}^4 + \sum_{\alpha < \beta} T_{\alpha\beta} (J_{\alpha}^2 J_{\beta}^2 + J_{\beta}^2 J_{\alpha}^2) = T_{xx} J_x^4 + T_{yy} J_y^4 + T_{zz} J_z^4 \\
 &\quad + T_{xy} (J_x^2 J_y^2 + J_y^2 J_x^2) + T_{xz} (J_x^2 J_z^2 + J_z^2 J_x^2) + T_{yz} (J_y^2 J_z^2 + J_z^2 J_y^2), \\
 \hat{H}_6 &= \sum_{\alpha} \Phi_{\alpha\alpha\alpha} J_{\alpha}^6 + \sum_{\alpha < \beta} \Phi_{\alpha\alpha\beta} (J_{\alpha}^4 J_{\beta}^2 + J_{\beta}^2 J_{\alpha}^4) + \Phi_{xyz} (J_x^2 J_y^2 J_z^2 + J_z^2 J_y^2 J_x^2),
 \end{aligned} \tag{5}$$

184 or in an equivalent and more handy cylindrical representation, where subscripts denote powers of
 185 $|\mathbf{J}|, J_z, J_{\pm}$, accordingly: $A_{klm}(J) = A_{klm} |\mathbf{J}|^k J_z^l (J_+^m + J_-^m)$:

$$\begin{aligned}
 H_{rot}^{(cyl)} &= H_{02}^{(cyl)} + H_{04}^{(cyl)} + H_{06}^{(cyl)} + H_{08}^{(cyl)} = B_{200} \mathbf{J}^2 + B_{020} J_z^2 + B_{002} (J_+^2 + J_-^2) \\
 &\quad + T_{400} (\mathbf{J}^2)^2 + T_{220} \mathbf{J}^2 J_z^2 + T_{040} J_z^4 + \frac{1}{2} \{ T_{202} \mathbf{J}^2 + T_{022} J_z^2, J_+^2 + J_-^2 \}_+ + T_{004} (J_+^4 + J_-^4) \\
 &\quad + \Phi_{600} (\mathbf{J}^2)^3 + \Phi_{420} (\mathbf{J}^2)^2 J_z^2 + \Phi_{240} \mathbf{J}^2 J_z^4 + \Phi_{060} J_z^6 + \frac{1}{2} \{ \Phi_{204} \mathbf{J}^2 + \Phi_{024} J_z^2, J_+^4 + J_-^4 \}_+ \\
 &\quad + \frac{1}{2} \{ \Phi_{402} (\mathbf{J}^2)^2 + \Phi_{222} \mathbf{J}^2 J_z^2 + \Phi_{042} J_z^4, J_+^2 + J_-^2 \}_+ + \Phi_{006} (J_+^6 + J_-^6).
 \end{aligned} \tag{6}$$

186 In order to transform the original vibration-rotation Hamiltonian operator $H \equiv H_{rot}^{(cyl)}$ to the
 187 required (quasi-)diagonal form, suitable for subsequent fine fitting to observed data for a selected
 188 vibrational state or polyad of states, it is customary to apply a sequence of n unitary similarity
 189 transformations (also, ‘contact’ transformations) $U_k = \exp(iS_k)$, which gradually reduce it to a
 190 desired shape [24, 30]:

$$H' = U H U^{-1} = U_n \dots U_2 U_1 H U_1^{-1} U_2^{-1} \dots U_n^{-1}, \tag{7}$$

191 where the exponential gains a practical form using the Backer-Campbell-Hausdorff (BCH) expan-
 192 sion:

$$\hat{H}' = \exp(iS) \cdot \hat{H} \cdot \exp(-iS) = \hat{H} + i[\hat{S}, \hat{H}] - \frac{1}{2!} [\hat{S}, [\hat{S}, \hat{H}]] - \frac{i}{3!} [\hat{S}, [\hat{S}, [\hat{S}, \hat{H}]]] + \dots \tag{8}$$

193 In practice, it is most common to use two contact transformations ($n = 2$) to obtain the desired
 194 ‘reduced’ new form of H' up to sextic constants, or, less frequently, three contact transformations
 195 were explicitly used for obtaining an analytic form of octic H' [31, 32].

196 Using our updated notation, the general CVPT expansion (7) reads:

$$H'_{02} = H_{02}, \tag{9a}$$

$$H'_{04} = H_{04} + i[S_3, H_{02}], \tag{9b}$$

$$H'_{06} = H_{06} - \frac{1}{2} [S_3, [S_3, H_{02}]] + i[S_3, H_{04}] + i[S_5, H_{02}].$$

197 After accomplishing these contact transformations the resulting Hamiltonian is merely a sum of
 198 transformed contributions:

$$\hat{H}'_{rot} = \hat{H}'_0 + \hat{H}'_1 + \hat{H}'_2 + \hat{H}'_3. \tag{10}$$

199 Sarka *et al.* [13] introduced the following order-of-magnitude condition for preserving the per-

200 turbative hierarchy under a contact transformation:

$$|s_{prt}| \lesssim \kappa^{2(p+r+t-1)}, \quad \kappa^4 \sim T/B, \quad |s_{111}| \lesssim \kappa^4 \sim T/B, \quad (11)$$

201 where T and B denote typical quartic centrifugal distortion and principal rotational constants,
202 respectively. For SO_2F_2 , the A -reduction gives

$$\left| s_{111}^{(A)} \right| \simeq 4 \times 10^{-5}, \quad \kappa^4 \simeq 5 \times 10^{-7}, \quad \frac{\left| s_{111}^{(A)} \right|}{\kappa^4} \sim 10^2. \quad (12)$$

203 Sarka *et al.* interpreted this excess as a loss of rapid convergence and as a possible divergence
204 of the truncated rotational Hamiltonian at medium J and K values. It should be emphasized,
205 however, that Eq. (11) is a power-counting criterion rather than a rigorous convergence boundary:
206 its violation indicates that the induced sextic and higher-power contributions no longer follow
207 their expected hierarchy.

208 The A -reduced sextic effective Watson Hamiltonian is given by [4, 23]:

$$\begin{aligned} H_{\text{rot}}^{(A)} &= H_{\text{quad}}^{(A)} + H_{\text{quartic}}^{(A)} + H_{\text{sextic}}^{(A)} \\ H_{\text{quad}}^{(A)} &= B_x^{(A)} J_x^2 + B_y^{(A)} J_y^2 + B_z^{(A)} J_z^2 \\ &= \frac{1}{2} (B_x^{(A)} + B_y^{(A)}) \mathbf{J}^2 + [B_z^{(A)} - \frac{1}{2} (B_x^{(A)} + B_y^{(A)})] J_z^2 + \frac{1}{4} (B_x^{(A)} - B_y^{(A)}) (J_+^2 + J_-^2), \\ H_{\text{quartic}}^{(A)} &= -\Delta_J (\mathbf{J}^2)^2 - \Delta_{JK} \mathbf{J}^2 J_z^2 - \Delta_K J_z^4 - \frac{1}{2} \{ \delta_J \mathbf{J}^2 + \delta_K J_z^2, J_+^2 + J_-^2 \}, \\ H_{\text{sextic}}^{(A)} &= \Phi_J (\mathbf{J}^2)^3 + \Phi_{JK} (\mathbf{J}^2)^2 J_z^2 + \Phi_{KJ} \mathbf{J}^2 J_z^4 + \Phi_K J_z^6 \\ &\quad + \frac{1}{2} \{ \phi_J (\mathbf{J}^2)^2 + \phi_{JK} \mathbf{J}^2 J_z^2 + \phi_K J_z^4, J_+^2 + J_-^2 \}. \end{aligned} \quad (13)$$

209 The necessity of using an alternative type of rotational reduction arises when an asymmetric
210 top occasionally becomes a quasi-prolate/oblate symmetric top [5–7, 33, 34]. It is achieved by
211 eliminating all J_z -dependent matrix elements for $|\Delta K| > 0$: $T'_{mnk}, \Phi'_{mnk}, \Theta'_{mnk} = 0$, where $n >$
212 $0, k > 0$.

$$\begin{aligned} \tilde{H}_{\text{rot}}^{(S)} &= B_x^{(S)} J_x^2 + B_y^{(S)} J_y^2 + B_z^{(S)} J_z^2 \\ &\quad - D_J (\mathbf{J}^2)^2 - D_{JK} \mathbf{J}^2 J_z^2 - D_K J_z^4 + d_1 \mathbf{J}^2 (J_+^2 + J_-^2) + d_2 (J_+^4 + J_-^4) \\ &\quad + H_J (\mathbf{J}^2)^3 + H_{JK} (\mathbf{J}^2)^2 J_z^2 + H_{KJ} \mathbf{J}^2 J_z^4 + H_K J_z^6 \\ &\quad + h_1 (\mathbf{J}^2)^2 (J_+^2 + J_-^2) + h_2 \mathbf{J}^2 (J_+^4 + J_-^4) + h_3 (J_+^6 + J_-^6). \end{aligned} \quad (14)$$

213 3. Discussion

214 In this Section we will consider a number of theoretical points, concerning the proofs of applica-
215 bility of the ‘6-reduction’, introduced by Sarka *et al.* At first, we will follow the Watson’s primary
216 argument of existence of only five linear combinations of six quartic constants T_{klm} , outlined on his
217 first publication [1]. In the first order, adding a commutator $i[\hat{S}_3, H_0]$ preserves the spectrum but
218 allows to cancel one redundant constant. The corresponding transformation of quartic constants
219 produces a *gauge vector* in the nine-parameter space of rotational constants A, B, C and cylindri-
220 cal quartic coefficients $T_{klm}, k+l+m=4$. Next, we prove that the gauge vector is a null-vector in
221 Jacobian matrix $\mathbb{J}$, and consequently in the inverse $(\mathbb{J}^\top \mathbf{W} \mathbb{J})^{-1}$ does not exist, which means that
222 only $\text{rank}(\mathbb{J}) \leq 8$ independent combinations of the three rotational and six quartic constants are
223 determinable from rotational spectra.

224 These mathematical facts have a deep underlying physical reasons, considered in Sec. 3.3.
225 We construct a semiclassical analogue of a distorted rotating molecule and show that the unitary

transformation $H' = \exp(i\hat{S}_3) \cdot H \cdot \exp(-i\hat{S}_3)$ can be considered as a J -dependent rotation of molecule-fixed frame that introduces an unavoidable ambiguity, called *gauge freedom*, that makes one quartic operator redundant as it is associated with a molecule, rotating as a whole body.

Next, we investigate the dependence between R_5 and R_6 and derive the first-order quartic spectroscopic invariant $R_5 + 2\sigma R_6 = C_{R_5 R_6}$ that remains constant along the gauge direction, defined by s_{111} . Since parameter $\sigma = (2B_z - B_x - B_y)/(B_x - B_y)$ divides quantities of the same order of magnitude, it remains numerically stable for SO_2F_2 and can serve as a criterion for choosing the type of reduction.

Subsection 3.5 investigates the consequences of evaluating the value of s_{111} in the second order of perturbation theory from approximate equations for P'_{006} . We provide a number of counterarguments against this kind of ‘delayed’ reduction, where the first order fixing of $\hat{S}_3$ is conducted using second order relationships.

The effect of using constraint $h_1 = 0$ on confidence intervals of the first-order quartic parameters in 6-reduction is discussed in Subsection 3.6. We provide an explanation that small values of obtained confidence intervals demonstrate the numerical stability of the fit within the selected reduction convention, but they cannot establish the independent physical determinability of all six quartic constants.

In the last Subsection we discuss main conclusions that follow from a significant number of post-Sarka *et al.* works. Many of them employ the irreducible tensor approach (ITO) formalism. Our analysis shows that the distinction between numerical determinability within a constrained model and genuine independence of spectroscopic constants remained unresolved.

3.1. First-Order Expectation Value and Gauge Vector

In his first brief letter to the Editor of J. Chem. Phys. [1], concerning the indeterminacy of the quartic rotational Hamiltonian, Watson utilized the fundamental consequence of the quantum mechanical hypervirial theorem, which states that for any general (e.g., Hermitian angular momentum) operator $\hat{O}$, the expectation value $\hat{C}_k$ of the commutator $[\hat{O}, H_0]$ is identically zero:

$$\begin{aligned}\hat{C}_k(\hat{O}) &= \langle \psi_k | [\hat{O}, H_0] | \psi_k \rangle = \langle \psi_k | \hat{O} \cdot H_0 | \psi_k \rangle - \langle \psi_k | H_0 \cdot \hat{O} | \psi_k \rangle \\ &= \langle \psi_k | \hat{O}(E_k | \psi_k \rangle) - (E_k \langle \psi_k |) \hat{O} | \psi_k \rangle = E_k (\langle \psi_k | \hat{O} | \psi_k \rangle - \langle \psi_k | \hat{O} | \psi_k \rangle) = 0.\end{aligned}\quad (15)$$

where $|\psi_k\rangle$ is an exact eigenfunction of the (e.g., rigid-rotor) Hamiltonian H_0 ($H_0 |\psi_k\rangle = E_k |\psi_k\rangle$).

Operator $\hat{O}$. Let us now consider a more specific form of $\hat{O}$ in the context of a quartic rotational Hamiltonian for an orthorhombic molecule. Since the zero-order Hamiltonian H_0 is of two in angular momentum, and the time-reversal symmetry of molecular rotation forbids operators of odd overall degree, we are restricted to generating even degree corrections, originating from $\hat{C}_k(\hat{O})$. An evaluation of $\hat{C}(\hat{O}^{(n)}) = [\hat{O}^{(n)}, H_0]$ produces an operator of degree $(n+2) - 1$, and therefore an allowed value of n is 3. The only cubic Cartesian form of $\hat{O}^{(3)}$, satisfying these requirements, is the symmetrized product of all three angular momentum components $(J_x^k J_y^l J_z^m + J_z^m J_y^l J_x^k)$, multiplied by an arbitrary scalar parameter o_{klm} . It can be immediately verified that a fully symmetric direct product $\Gamma(\hat{O}^{(n)}) = \Gamma(J_x^k) \otimes \Gamma(J_y^l) \otimes \Gamma(J_z^m)$ with the condition $k + l + m = 3$ for the rotational point symmetry group D_2 is fully symmetric if and only if when $k = l = m = 1$.

Operator $\hat{S}$. For historical reasons and in the context of rotational perturbation theory the zero-order operator $H_0 \equiv H_2$ and the cubic operator, described above, is denoted as follows:

$$\hat{S}_3 = s_{111}(J_x J_y J_z + J_z J_y J_x). \quad (16)$$

Next, if the rotational Hamiltonian is subjected to the unitary transformation, the first-order correction is given by the commutator $i[\hat{S}_3, H_0]$:

$$\hat{H}' = \exp(i\hat{S}_3) \cdot \hat{H} \cdot \exp(-i\hat{S}_3) = \hat{H} + i[\hat{S}_3, H_0] + O(\hat{J}^6). \quad (17)$$

Now we can draw a conclusion that the expectation value of the quartic centrifugal distortion operator H_{rot} with added commutator $i[\hat{S}_3, H_0]$ is independent of a value of scalar parameter s_{111} :

$$\langle \psi_k | H_0 + H_4 | \psi_k \rangle = \langle \psi_k | H_0 + H_4 + i[s_{111}(J_x J_y J_z + J_z J_y J_x), H_0] | \psi_k \rangle = \langle \psi_k | H'_{\text{rot}} | \psi_k \rangle. \quad (18)$$

where $|\psi_k\rangle$ are the H_0 wave functions. Indeed, since $\langle \psi_k | i[s_{111}(J_x J_y J_z + J_z J_y J_x), H_0] | \psi_k \rangle \equiv 0$, the term containing s_{111} vanishes upon averaging. Therefore, $\langle H'_{\text{rot}} \rangle = \langle H_{\text{rot}} \rangle$ for any arbitrary value of s_{111} . This independence means that in the first order the rotational energy of a quartic centrifugal distortion Hamiltonian H_{rot} is invariant of the certain operator transformation that modifies separate terms in the consistent way. In other words, individual coefficients in full original form of H_{rot} are not physically determinable, and in order to make the Hamiltonian only consisting of determinable terms, it is necessary to choose a specific value of s_{111} which removes the linear dependency. Analytically, the commutator $i[\hat{S}_3, H_0]$ in (18) evaluates to a specific mixture of quadratic and quartic operators [1]:

$$i[s_{111}(J_x J_y J_z + J_z J_y J_x), H_0] = 2s_{111} \{ (B_x - B_y)(J_x^2 J_y^2 + J_y^2 J_x^2 + 2J_z^2) + (B_y - B_z)(J_y^2 J_z^2 + J_z^2 J_y^2 + 2J_x^2) + (B_z - B_x)(J_z^2 J_x^2 + J_x^2 J_z^2 + 2J_y^2) \}. \quad (19)$$

Applying the hypervirial theorem (15), we find that the expectation value of the left side of (19) is exactly zero. Consequently, introducing this commutator into the Hamiltonian modifies both quadratic and quartic constants, but this operation does not linearly alter the rotational energy spectrum.

It can be seen from (18), (19) that the expectation value of the transformed Hamiltonian H'_{rot} :

$$\hat{H}'_{\text{rot}} = \hat{H}'_0 + \hat{H}'_1 = \sum_{\alpha} B'_{\alpha} J_{\alpha}^2 + \sum_{\alpha} T_{\alpha} J_{\alpha}^4 + \sum_{\alpha < \beta} T'_{\alpha\beta} (J_{\alpha}^2 J_{\beta}^2 + J_{\beta}^2 J_{\alpha}^2). \quad (20)$$

is exactly the same as the original full Hamiltonian H_{rot} (21), where:

$$\begin{aligned} B'_x &= B_x + 4s_{111}(B_y - B_z); & B'_y &= B_y + 4s_{111}(B_z - B_x); & B'_z &= B_z + 4s_{111}(B_x - B_y), \\ T'_{xy} &= T_{xy} + 2s_{111}(B_x - B_y); & T'_{xz} &= T_{xz} + 2s_{111}(B_z - B_x); & T'_{yz} &= T_{yz} + 2s_{111}(B_y - B_z). \end{aligned} \quad (21)$$

An equivalent and more convenient cylindrical representation, where subscripts denote powers of $|\mathbf{J}|, J_z, J_{\pm}$: $A_{klm}(J) = A_{klm} |\mathbf{J}|^k J_z^l (J_+^m + J_-^m)$, reads:

$$\begin{aligned} B'_{200} &= B_{200} + 4s_{111}B_{002}, & B'_{020} &= B_{020} - 14s_{111}B_{002}, & B'_{002} &= B_{002} + 2s_{111}B_{020}, \\ T'_{400} &= T_{400} - 2s_{111}B_{002}, & T'_{220} &= T_{220} + 12s_{111}B_{002}, & T'_{040} &= T_{040} - 10s_{111}B_{002}, \\ T'_{202} &= T_{202}, & T'_{022} &= T_{022} - 2s_{111}B_{020}, & T'_{004} &= T_{004} + s_{111}B_{002}. \end{aligned} \quad (22)$$

Such s_{111} -controlled correlated changes in quadratic and quartic terms do not alter the energy in the first order, or, first derivatives of individual rotational energy levels are zero: $\partial E_m / \partial s_{111} = 0$. This means that transformed parameters can be shifted continuously along a one-dimensional direction ('*gauge orbit*'), defined by s_{111} , and this change has no physical effect. This mathematical degree of freedom, arising due to arbitrariness of parameter s_{111} , may be interpreted as a rotational *gauge freedom*, associated with the choice of the molecule-fixed frame [1, 35].

292 Gauge freedom is the existence of a non-trivial direction $\mathbf{g}^9$ in the parameter space $\mathbf{P} =$
 293 $(A, B, C, T_{400}, \dots, T_{004})$ such that $\mathbf{P}' = \mathbf{P} + s_{111} \mathbf{g}^9$ leaves all observable rotational energies un-
 294 changed if higher-order corrections to energy can be neglected:

$$E_n^{(2+4)}(\mathbf{P}') - E_n^{(2+4)}(\mathbf{P}) = \mathcal{O}(J^6). \quad (23)$$

295 Sextic-order terms may produce curvature along a quartic gauge direction in a truncated parametriza-
 296 tion, but that curvature is representation- and truncation-dependent unless all terms participating
 297 in the corresponding higher-order contact transformation are included consistently.

298 The system of equations (22) can be formulated in terms of the so-called nine-dimensional
 299 ‘gauge-vector’:

$$\mathbf{g}^9 = \{4B_{002}, -14B_{002}, +2B_{020}, -2B_{002}, 12B_{002}, -10B_{002}, 0, -2B_{020}, B_{002}\} \in \mathbb{R}^9. \quad (24)$$

300 This vector can be added to the original set to form $\mathbf{P}' = \mathbf{P} + s_{111} \mathbf{g}^9$, and such operation can be
 301 called ‘*gauge transformation*’ that defines a null vector in the entire space $\mathbf{P} = \mathbf{B} \cup \mathbf{T}$, formed
 302 by parameters B_k, T_l . It can be noticed that components of $\mathbf{g}^9$ are given by first derivatives
 303 $\partial B_k / \partial s_{111}, \partial T_l / \partial s_{111}$, and this fact will be taken advantage of in the next Section, concerning
 304 least-squares procedure.

Table 1: Transformation of quartic centrifugal-distortion constants under the s_{111} contact transformation and gauge vector $\mathbf{g}$.

Reduced	A_{klm} coefficient	s_{111} -dependent term	Abbreviation	$\mathbf{g} = \partial A'_{klm} / \partial s_{111}$
B'_{200}	B_{200}	$s_{111}(B_x - B_y)$		Δ_{xy}
B'_{020}	B_{020}	$-\frac{7}{2}s_{111}(B_x - B_y)$		$-\frac{7}{2}\Delta_{xy}$
B'_{002}	B_{002}	$s_{111}(2B_z - B_x - B_y)$		Λ_{xyz}
T'_{400}	T_{400}	$-\frac{1}{2}s_{111}(B_x - B_y)$	$-D_J$	$-\frac{1}{2}\Delta_{xy}$
T'_{220}	T_{220}	$3s_{111}(B_x - B_y)$	$-D_{JK}$	$3\Delta_{xy}$
T'_{040}	T_{040}	$-\frac{5}{2}s_{111}(B_x - B_y)$	$-D_K$	$-\frac{5}{2}\Delta_{xy}$
T'_{202}	T_{202}	0	$-\delta_J$	0
T'_{022}	T_{022}	$-s_{111}(2B_z - B_x - B_y)$	$2R_5$	$-\Lambda_{xyz}$
T'_{004}	T_{004}	$\frac{1}{4}s_{111}(B_x - B_y)$	R_6	$\frac{1}{4}\Delta_{xy}$

$$\Delta_{xy} = B_x - B_y, \quad \Lambda_{xyz} = 2B_z - B_x - B_y.$$

305 Equations (21),(22) prove that a simultaneous, correlated change in the quadratic rigid-rotor
 306 constants and the quartic centrifugal distortion cross-terms, $(B_\alpha, T_{\alpha\beta} \rightarrow B'_\alpha, T'_{\alpha\beta})$ leaves the first-
 307 order energy invariant. If higher-order corrections to energy can be neglected, it is fundamentally
 308 impossible to distinguish between the original parameter set and the transformed form using the
 309 observed transition frequencies. The vector $\mathbf{g}^9$ can be interpreted as the direction in parameter
 310 space that corresponds to J -dependent rotation of the coordinate frame rather than physically
 311 stretching the molecule. Before we will consider this interpretation in detail, let us consider direct
 312 consequences of existence of the gauge vector $\mathbf{g}^9$ to building a least squares procedure for fitting
 313 spectroscopic parameters.

3.2. Gauge Vector and Least Squares Method (LSQ)

For performing a least-squares fit of rotational energy terms or transitions, it is necessary to compose the $[n \times p]$ Jacobian matrix $\mathbb{J} = \partial E_\ell / \partial p_j$ or $\mathbb{J} = \partial \nu_i / \partial p_j$, where rows correspond to measured empirical energy levels E_ℓ or transition frequencies $\nu_i = (E'_{\ell'} - E''_{\ell''})$ and nine columns, corresponding to spectroscopic constants A_k : $B_j, j = 1 \dots 3$ and $T_j, j = 1 \dots 6$. Without loss of generality, we will choose transition frequencies ν_i in capacity of empirical data in the context of LSQ procedure. Since the derivatives of both lower and upper energy states with respect to s_{111} are exactly zero, for any transition $\partial(E'_{\ell'} - E''_{\ell''}) / \partial s_{111} = \partial \nu_i / \partial s_{111} = 0$. Applying the chain rule, we obtain the following scalar product condition:

$$\frac{\partial \nu_i}{\partial s_{111}} = \sum_{j=1}^9 \frac{\partial \nu_i}{\partial A_j} \cdot \frac{\partial A_j}{\partial s_{111}} = \mathbb{J}_i \cdot \mathbf{g} = 0. \quad (25)$$

Since this holds for every row $\mathbb{J}_i$, the full matrix equation is given by:

$$\begin{pmatrix} \frac{\partial \nu_1}{\partial B_{200}} & \frac{\partial \nu_1}{\partial B_{020}} & \frac{\partial \nu_1}{\partial B_{002}} & \frac{\partial \nu_1}{\partial T_{400}} & \cdots & \frac{\partial \nu_1}{\partial T_{004}} \\ \frac{\partial \nu_2}{\partial B_{200}} & \frac{\partial \nu_2}{\partial B_{020}} & \frac{\partial \nu_2}{\partial B_{002}} & \frac{\partial \nu_2}{\partial T_{400}} & \cdots & \frac{\partial \nu_2}{\partial T_{004}} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \frac{\partial \nu_n}{\partial B_{200}} & \frac{\partial \nu_n}{\partial B_{020}} & \frac{\partial \nu_n}{\partial B_{002}} & \frac{\partial \nu_n}{\partial T_{400}} & \cdots & \frac{\partial \nu_n}{\partial T_{004}} \end{pmatrix} \begin{pmatrix} g_1 \\ g_2 \\ g_3 \\ g_4 \\ \vdots \\ g_9 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}. \quad (26)$$

This equality means that $\mathbf{g}$ is a null-vector for matrix $\mathbb{J}$.

In practical LSQ calculations, an actual symmetrized system of normal equations (Gram matrix) usually contains the diagonal weight matrix $\mathbf{W}$ and is given by:

$$\mathbf{G} \mathbf{g} = \mathbb{J}^\top \mathbf{W} \mathbb{J} \mathbf{g} = \mathbb{J}^\top \mathbf{W} (\mathbb{J} \mathbf{g}) = \mathbf{0}. \quad (27)$$

Hence, $\mathbf{g}$ is an exact eigenvector of the Gram matrix $\mathbf{G}$ with zero eigenvalue, and therefore:

$$\det(\mathbb{J}^\top \mathbf{W} \mathbb{J}) = 0. \quad (28)$$

The inverse $(\mathbb{J}^\top \mathbf{W} \mathbb{J})^{-1}$ does not exist, and consequently only $\text{rank}(\mathbb{J}) \leq 8$ independent combinations of the three rotational and six quartic constants are determinable from rotational spectra. The ratio $\eta = \Sigma_{\max} / \Sigma_{\min}$ of maximum and minimum singular values of matrix $\mathbf{W}^{1/2} \mathbb{J} = V \Sigma U$ can serve as a measure of ill-conditioning of the LSQ problem. This criterion can be supplemented by a comparison of s_{111} values, calculated *ab initio* for different representations ($A, B, C \leftrightarrow \alpha, \beta, \gamma = \{x, y, z\}$), to form a reliable estimation of correctly posed LSQ problem [36].

So far, we have considered the first-order linear dependence among the rotational and quartic centrifugal-distortion constants. This construction is exact through the retained quartic order. At the second order, the perturbation theory expansion generates contributions from $i[S_3, H_4]$ and from the second nested commutator $-\frac{1}{2}[S_3, [S_3, H_0]]$. We now consider the semiclassical interpretation of the quartic null direction.

3.3. Semiclassical Quartic Hamiltonian and the Gauge Freedom

In the preceding two Sections, we have shown that the quantum contact transformation establishes the unitary equivalence of the family of rotational Hamiltonians $H' = \exp(-i\hat{S}) \cdot H \cdot \exp(i\hat{S})$ and hence preserves their spectra, while revealing the redundancy of one quartic coefficient in

the first order. Within this quantum picture, the molecule-fixed angular momentum components $\hat{J}_\alpha$ do not commute and cannot be identified with the components of a single classical phase space vector. Therefore, a geometrical interpretation of the unitary transformation is not directly accessible.

The usefulness of a semiclassical formulation in high-order rotational theory was explicitly recognized and actively exploited by Watson [31]. He represented the vibration-rotation Hamiltonian as a polynomial in the classical variables $\mathcal{Q}$, $\mathcal{P}$, and $\mathcal{J}$, and expressed the successive contact transformations through Poisson brackets $\{\mathcal{S}, \mathcal{H}\} \approx -i[S, H]$. Of particular importance here, the angular momentum $\mathcal{J}$ was transformed together with $\mathcal{Q}$ and $\mathcal{P}$, thereby preserving the canonical character of the complete transformation before the vibrational variables were set to zero. This classical calculation reproduced the known quartic and sextic centrifugal distortion terms and yielded the same octic Hamilton function $\tilde{H}_{08}$ as the corresponding quantum perturbation treatment. In practical calculations, Watson evaluated the unreduced rotational Hamilton function for a set of directions of $\mathcal{J}$ and recovered its homogeneous polynomial coefficients by least squares. More generally, classical limits of molecular effective Hamiltonians and rotational-energy surfaces have been successfully used to visualize the body-fixed angular momentum dynamics and to interpret the structure of high- J rotational spectra [37–39].

In this Section we will show that in the semiclassical limit, the first order contact transformation represents a momentum-dependent instantaneous reorientation of the molecule-fixed frame. This freedom introduces an intrinsic ambiguity in the geometrical interpretation of one component of the quartic centrifugal deformation: its contribution may be represented as a molecular distortion in a fixed frame or absorbed into a momentum-dependent reorientation of that frame, while the rotational spectrum remains unchanged to the retained perturbative accuracy [31].

In classical angular momentum phase space, the counterpart of the quantum contact transformation is a s_{111} -parameterized canonical flow¹ that can be represented in the first order by the momentum-dependent 3×3 rotational matrix $\mathcal{R}(\mathcal{J})$, describing an infinitesimal reorientation of the effective molecule-fixed frame. Under the principal-symbol mapping,² the operators $\hat{J}_\alpha$ are represented by the commuting variables $\mathcal{J}_\alpha$, while the leading commutator contribution is given by the corresponding Lie-Poisson bracket $\{\mathcal{H}_0, \mathcal{S}_3\}_{\text{LP}}$. This principal symbol approximation to $i[\hat{S}_3, \hat{H}_2]$ correctly reproduces the quartic part of the transformation but omits the lower degree quadratic terms, generated by the angular momentum commutation relations; these terms produce correlated changes in the effective rotational constants. Nevertheless, the semiclassical construction provides a geometrical visualization of the gauge freedom, whereas exact isospectrality remains a consequence of the quantum unitary transformation.

Classical Quartic Hamiltonian. In the standard quantum approach, the Watson vibration-rotation Hamiltonian $\hat{H}_{\text{vib-rot}}$ is subjected to vibrational contact transformations and subsequently averaged over a selected vibrational state v [24]. The resulting state-dependent effective rotational Hamiltonian $H_{\text{rot}}(v)$ is further reduced using a series of rotational unitary transformations [23, 24, 27, 31].

¹A canonical flow is a one-parameter family of phase-space transformations, generated by a phase-space function through the Poisson bracket. It preserves the Lie-Poisson structure and its Casimir operator $\mathcal{J}^2$.

²A semiclassical symbol $\mathcal{A}(z; \hbar)$ is a phase space function assigned to a quantum operator $\hat{A}$:

$$\hat{A} = \text{Op}_\hbar[\mathcal{A}], \quad \mathcal{A}(z; \hbar) \sim \mathcal{A}_0(z) + \hbar \mathcal{A}_1(z) + \hbar^2 \mathcal{A}_2(z) + \dots$$

Here, the principal symbol $\mathcal{A}_0(z)$ is the leading term; the remaining terms contain quantum corrections, including operator-ordering effects. The correspondence between a quantum commutator and its Lie-Poisson bracket is:

$$\text{symb}(-i\hbar[\hat{A}, \hat{B}]) = \{A_0, B_0\}_{\text{LP}} + \text{quantum corrections.}$$

381 To construct its semiclassical counterpart, let $\mathbf{Q} = (Q_1, \dots, Q_f)$ and $\mathbf{P} = (P_1, \dots, P_f)$ de-
 382 note mass-weighted vibrational normal coordinates and their conjugate momenta, while $\mathbf{J} =$
 383 (J_x, J_y, J_z) denotes the angular momentum in a molecule-fixed Eckart frame [35, 40]. Expansion
 384 of the corresponding semiclassical Hamiltonian yields the following rotation-dependent zero-, first-
 385 and second-order contributions:

$$\mathcal{H}_{\text{rot}}^{(2)} = B_x J_x^2 + B_y J_y^2 + B_z J_z^2. \quad (29)$$

$$\mathcal{H}_{\text{vib-rot}}^{(3)} = \frac{1}{2} \sum_{\alpha\beta r} \mu_{\alpha\beta,r}^{(1)} Q_r J_\alpha J_\beta - \sum_{\alpha\beta} \mu_{\alpha\beta}^{(0)} J_\alpha \pi_\beta, \quad (30)$$

$$\mathcal{H}_{\text{vib-rot}}^{(4)} = \frac{1}{4} \sum_{\alpha\beta rs} \mu_{\alpha\beta,rs}^{(2)} Q_r Q_s J_\alpha J_\beta - \sum_{\alpha\beta r} \mu_{\alpha\beta,r}^{(1)} J_\alpha Q_r \pi_\beta. \quad (31)$$

388 where the reciprocal inertia tensor, expanded in the equilibrium configuration, reads:

$$\mu_{\alpha\beta}(\mathbf{Q}) = \mu_{\alpha\beta}^{(0)} + \sum_r \mu_{\alpha\beta,r}^{(1)} Q_r + \frac{1}{2} \sum_{rs} \mu_{\alpha\beta,rs}^{(2)} Q_r Q_s + \dots, \quad \mu_{\alpha\beta}^{(0)} = 2B_\alpha \delta_{\alpha\beta}. \quad (32)$$

389 while $\boldsymbol{\pi}$ is the vibrational angular momentum: $\pi_\alpha = \sum_{rs} \zeta_{rs}^\alpha Q_r P_s$.

390 The semiclassical analogue of the second order quantum vibrational contact transformations
 391 is obtained by introducing two canonical Lie transformations [30, 31],

$$\mathcal{H}'_{\text{vib-rot}}(\mathbf{Q}, \mathbf{P}, \mathbf{J}) = \exp(\mathcal{L}_{\mathcal{S}_2}) \cdot \exp(\mathcal{L}_{\mathcal{S}_1}) \cdot \mathcal{H}, \quad \mathcal{L}_{\mathcal{S}_k} \cdot \mathcal{H} = \{\mathcal{H}, \mathcal{S}_k\}_{\text{vib-rot}}. \quad (33)$$

392 Here, $\mathcal{S}_1$ and $\mathcal{S}_2$ are the semiclassical symbols of the vibrational contact transformation generators,
 393 which remove the nonresonant terms, depending on the vibrational coordinates and momenta [24].
 394 In Eq. (33), the complete vibration-rotation Poisson bracket is given by [31, 41]:

$$\{\mathcal{S}, \mathcal{H}\}_{\text{vib-rot}} = \sum_r \left(\frac{\partial \mathcal{S}}{\partial Q_r} \frac{\partial \mathcal{H}}{\partial P_r} - \frac{\partial \mathcal{S}}{\partial P_r} \frac{\partial \mathcal{H}}{\partial Q_r} \right) + \{\mathcal{S}, \mathcal{H}\}_{\text{LP}}, \quad (34)$$

395 where the molecule-fixed Lie-Poisson part $\{\cdot, \cdot\}_{\text{LP}}$ is defined below.

396 A semiclassical analogue of averaging over a selected vibrational state v can be constructed by
 397 representing its quantum wavefunction ψ_v by a normalized Wigner quasidistribution $\mathcal{W}_v(\mathbf{Q}, \mathbf{P})$
 398 in vibrational phase space [42, 43]. For harmonic normal modes, the ground-state Wigner qua-
 399 sidistribution $\mathcal{W}_0$ has the Gaussian form:

$$\mathcal{W}_0 = \prod_{r=1}^M \frac{1}{\pi \hbar} \exp \left[-\frac{1}{\hbar} \left(\omega_r Q_r^2 + \frac{P_r^2}{\omega_r} \right) \right]. \quad (35)$$

400 Interpreting $\mathcal{H}'_{\text{vib-rot}}$ as the leading Weyl symbol of the quasi-diagonal quantum Hamiltonian, the
 401 vibrational state rotational Hamilton function is defined by [43–45]:

$$\mathcal{H}_{\text{rot}}^{(v)}(\mathbf{J}) = \int \mathcal{W}_v(\mathbf{Q}, \mathbf{P}) \mathcal{H}'(\mathbf{Q}, \mathbf{P}, \mathbf{J}) d\mathbf{Q} d\mathbf{P}, \quad \int \mathcal{W}_v d\mathbf{Q} d\mathbf{P} = 1. \quad (36)$$

402 For the complete Weyl symbol, Eq. (36) reproduces $\langle \psi_v | \hat{H}' | \psi_v \rangle$ exactly. Replacement of the
 403 Weyl symbol by its leading classical part is a semiclassical approximation, and operator-ordering
 404 corrections must be considered separately when returning to the quantum operator form [45].

The resulting vibrational state effective rotational Hamilton function is given by:

$$\mathcal{H}_{\text{rot}}^{(v)}(\mathcal{J}) = \mathcal{H}_2(\mathcal{J}) + \mathcal{H}_4(\mathcal{J}) + \mathcal{O}(\mathcal{J}^6), \quad (37)$$

which can be subjected to the phase space transformation, generated by a classical function through Poisson brackets. For functions of the body-fixed angular momentum, the relevant bracket is the molecule-fixed Lie–Poisson bracket [31, 41, 46]

$$\{\mathcal{S}, \mathcal{H}\}_{\text{LP}} = -\mathcal{J} \cdot (\nabla_{\mathcal{J}} \mathcal{S} \times \nabla_{\mathcal{J}} \mathcal{H}), \quad \{\mathcal{J}_\alpha, \mathcal{J}_\beta\}_{\text{LP}} = -\varepsilon_{\alpha\beta\gamma} \mathcal{J}_\gamma. \quad (38)$$

Then a one-parameter rotational canonical transformation, generated by $\mathcal{S}$ can be written as

$$\mathcal{H}' = \exp(\mathcal{L}_{\mathcal{S}}) \cdot \mathcal{H}, \quad \mathcal{L}_{\mathcal{S}} \cdot \mathcal{H} = \{\mathcal{S}, \mathcal{H}\}_{\text{LP}}. \quad (39)$$

At the rotational power $\mathcal{J}^n$, reduction leaves $n + 1$ coefficients for an asymmetric top, including the orthorhombic case. Thus, five quartic, seven sextic, and nine octic coefficients remain after the corresponding rotational contact transformations [31]. The transformation maps phase space points, energy surfaces, and classical trajectories into their canonically equivalent representations, parameterized by the scalar multiplier s_{111} in $\mathcal{S}$.

The first canonical transformation can be accomplished using the following form of $\mathcal{S}$, that must be cubic in $\mathcal{J}$ and contain the factor of two to match the leading semiclassical symbol of the symmetrized quantum generator:

$$\mathcal{S}_3 = 2s_{111} \mathcal{J}_x \mathcal{J}_y \mathcal{J}_z. \quad (40)$$

This form preserves the Lie–Poisson bracket and the Casimir $\mathcal{J}^2$, so that it moves each angular-momentum point on the sphere $\mathcal{J}^2 = \text{const}$ without changing the underlying classical dynamics [41, 46]. For fixed $|\mathcal{J}|$, this sphere is the reduced classical phase space of molecular rotation [39, 47]. Applying the canonical transformation, specified in Eq. (39), to the quartic Hamilton function $\mathcal{H}_{\text{rot}}^{(v)}(\mathcal{J})$ yields:

$$\mathcal{H}'_{\text{rot}}(\mathcal{J}) = \mathcal{H}_2(\mathcal{J}) + \mathcal{H}_4(\mathcal{J}) + \{\mathcal{H}_2, \mathcal{S}_3\}_{\text{LP}} + \{\mathcal{H}_4, \mathcal{S}_3\}_{\text{LP}} + \frac{1}{2} \{ \{ \mathcal{H}_2, \mathcal{S}_3 \}_{\text{LP}}, \mathcal{S}_3 \}_{\text{LP}} + \cdots \quad (41)$$

Watson (2006) [31] explicitly constructed the reduced Hamiltonian by successive powers of $\mathcal{J}$ and rotational generators. Retaining the term linear in s_{111} , direct evaluation gives

$$\{\mathcal{H}_2, \mathcal{S}_3\}_{\text{LP}} = 4s_{111} [(B_x - B_y) \mathcal{J}_x^2 \mathcal{J}_y^2 + (B_y - B_z) \mathcal{J}_y^2 \mathcal{J}_z^2 + (B_z - B_x) \mathcal{J}_z^2 \mathcal{J}_x^2]. \quad (42)$$

This expression is the quartic principal symbol of the corresponding quantum commutator in Eq. (19). The additional terms, containing lower powers of $\hat{\mathcal{J}}$ in the quantum expression, arise from angular-momentum non-commutativity and have no counterpart here.

The geometrical meaning of Eq. (42) becomes evident by applying the same canonical transformation to the body-fixed angular-momentum components:

$$\delta \mathcal{J} = \{\mathcal{J}, \mathcal{S}_3\}_{\text{LP}} = \mathcal{J} \times \delta \boldsymbol{\vartheta}, \quad \delta \boldsymbol{\vartheta} = \nabla_{\mathcal{J}} \mathcal{S}_3 = 2s_{111} (\mathcal{J}_y \mathcal{J}_z, \mathcal{J}_x \mathcal{J}_z, \mathcal{J}_x \mathcal{J}_y). \quad (43)$$

Retaining terms linear in s_{111} , this transformation can be written as

$$\mathcal{J}' = \mathcal{R}(\mathcal{J}) \mathcal{J} + \mathcal{O}(s_{111}^2), \quad \mathcal{R}(\mathcal{J}) = \mathbf{I} - [\delta \boldsymbol{\vartheta}]_{\times} + \mathcal{O}(s_{111}^2), \quad (44)$$

where $[\mathbf{a}]_{\times} \mathbf{b} = \mathbf{a} \times \mathbf{b}$. Explicitly,

$$\mathcal{R}(\mathcal{J}) = \begin{pmatrix} 1 & \delta\vartheta_z & -\delta\vartheta_y \\ -\delta\vartheta_z & 1 & \delta\vartheta_x \\ \delta\vartheta_y & -\delta\vartheta_x & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2s_{111}\mathcal{J}_x\mathcal{J}_y & -2s_{111}\mathcal{J}_x\mathcal{J}_z \\ -2s_{111}\mathcal{J}_x\mathcal{J}_y & 1 & 2s_{111}\mathcal{J}_y\mathcal{J}_z \\ 2s_{111}\mathcal{J}_x\mathcal{J}_z & -2s_{111}\mathcal{J}_y\mathcal{J}_z & 1 \end{pmatrix} + \mathcal{O}(s_{111}^2). \quad (45)$$

The antisymmetric part of this matrix satisfies

$$\mathcal{R}^T(\mathcal{J})\mathcal{R}(\mathcal{J}) = \mathbf{I} + \mathcal{O}(s_{111}^2), \quad \det \mathcal{R}(\mathcal{J}) = 1 + \mathcal{O}(s_{111}^2). \quad (46)$$

Thus, locally and through terms linear in s_{111} , $\mathcal{R}(\mathcal{J})$ has the form of an infinitesimal rotation matrix. Because $\mathcal{R} = \mathbf{I} - [\delta\boldsymbol{\vartheta}]_{\times}$, it represents the passive change of vector components produced by a reorientation of the body-fixed coordinate frame through $\delta\boldsymbol{\vartheta}$.

Most importantly, substitution of the transformed components into the quadratic rigid rotor Hamilton function gives

$$\begin{aligned} \mathcal{H}_2(\mathcal{J}') - \mathcal{H}_2(\mathcal{J}) &= \nabla_{\mathcal{J}}\mathcal{H}_2 \cdot (\mathcal{J} \times \nabla_{\mathcal{J}}\mathcal{S}_3) + \mathcal{O}(s_{111}^2) \\ &= \{\mathcal{H}_2, \mathcal{S}_3\}_{\text{LP}} + \mathcal{O}(s_{111}^2). \end{aligned} \quad (47)$$

The right-hand side is precisely the quartic polynomial in Eq. (42). Consequently, the added quartic terms can be interpreted semiclassically as the change of the quadratic rotational energy produced by a momentum-dependent reorientation of the body-fixed frame. This establishes the connection between Eqs. (41) and (44).

For rotation about one principal axis, two components of $\mathcal{J}$ vanish and $\mathcal{R}(\mathcal{J})$ reduces to the identity matrix. When $\mathcal{J}_x$ and $\mathcal{J}_y$ are simultaneously nonzero, the local reorientation contains a rotation about z ,

$$\delta\vartheta_z(t) = 2s_{111}\mathcal{J}_x(t)\mathcal{J}_y(t). \quad (48)$$

Along a classical trajectory, this produces the rocking contact grid illustrated in panels (c) and (d). The instantaneous physical principal-axis frame is determined independently by the inertia tensor, as specified below. The physical principal-axis angle is fixed by the instantaneous inertia tensor, whereas the contact-frame angle is controlled by the freely chosen parameter s_{111} . Individual values of R_6 and the apparent grid tilt therefore depend on the contact convention, while the complete gauge-equivalence class carries the spectroscopic information.

This apparent motion is a representation-dependent reorientation of the coordinate grid and does not constitute an additional physical rotation. Accordingly, panels (c) and (d) of Figure 1 represent the same centrifugally deformed configuration in a fixed and a momentum-dependently reoriented effective frame, respectively.

Geometrical Illustration. Figure 1 summarizes the distinction between a physical centrifugal deformation and its representation in different contact gauges. A classical trajectory with $|\mathcal{J}|/\hbar \gg 1$ provides a useful semiclassical picture of the centrifugal deformation of a rotating molecule. We use planar SO_2 , lying in the xy plane with its C_2 axis along y , as a transparent illustration. The conclusions, that we will make will be equally applicable to SO_2F_2 .

In the underlying torque-free rigid-body motion, rotation is stable near the axes associated with the largest and smallest principal moments of inertia, whereas rotation about the intermediate axis is unstable [48]. Along a general stable torque-free trajectory, the molecule-fixed components $\mathcal{J}_x(t)$, $\mathcal{J}_y(t)$, and $\mathcal{J}_z(t)$ vary periodically with time t [49]. Quasiperiodic variation may arise when centrifugal deformation or additional vibration-rotation couplings are included. Consequently, both $\boldsymbol{\omega}(t)$ and $\delta\boldsymbol{\vartheta}[\mathcal{J}(t)]$ vary along the trajectory.

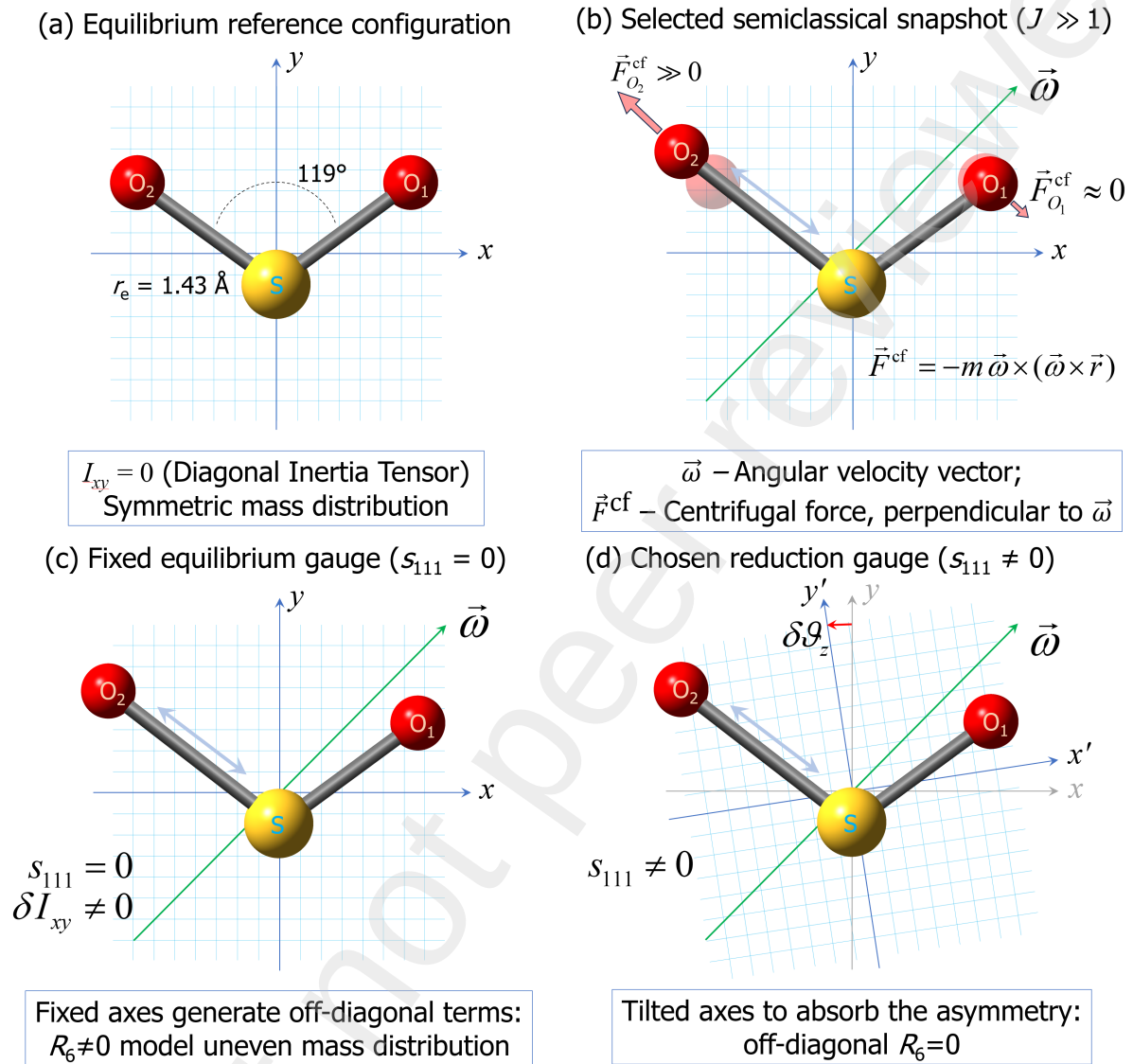


Figure 1: Semiclassical geometrical interpretation of the quartic contact-gauge freedom. Panel (a) shows the equilibrium SO_2 reference configuration and its equilibrium principal axes. Panel (b) presents a selected large-angular-momentum snapshot illustrating an asymmetric centrifugal force and the associated instantaneous deformation. The actual deformation is determined by the complete molecular force field; the geometry shown is only schematic. Panels (c) and (d) describe the same semiclassical rotational response in two contact gauges. In panel (c), the contact frame remains fixed, $s_{111} = 0$, and the selected quartic cross contribution is retained as $R_6 \neq 0$. In panel (d), the contact axes are locally reoriented by $\delta \vartheta_z = 2s_{111}J_x J_y$, with s_{111} chosen so that $R'_6 = 0$. The tilted grid represents a contact-frame convention and is not, in general, the instantaneous physical principal-axis system. At the leading semiclassical level, the transformation redistributes the quartic coefficients without changing B_α ; the corresponding quantum transformation also generates lower-power contributions through angular-momentum noncommutativity.

Under centrifugal forces, the corresponding displaced nuclear positions may be written schematically as [31, 50]:

$$\mathbf{a}'_i[\mathcal{J}(t)] = \mathbf{a}_i + \delta\mathbf{a}_i[\mathcal{J}(t)], \quad (49)$$

where $\mathbf{a}_i$ is equilibrium geometry atom coordinates, and $\delta\mathbf{a}_i$ represents its time-dependent change. In an adiabatic semiclassical treatment, the physical displacement can be obtained by minimizing the complete $\mathcal{J}$ -dependent effective potential [51].

Panel (a) defines the equilibrium reference configuration, whereas panel (b) presents one selected snapshot of this trajectory. Let the instantaneous angular velocity vector $\boldsymbol{\omega}(t)$ pass close to the S-O₁ bond. The centrifugal contribution acting on atom i is

$$\mathbf{F}_i^{(\text{cf})} = -m_i\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}_i) = m_i\boldsymbol{\omega} \times (\mathbf{r}_i \times \boldsymbol{\omega}), \quad (50)$$

and, for the geometry shown in panel (b),

$$|\mathbf{F}_{O_2}^{(\text{cf})}| \gg |\mathbf{F}_{O_1}^{(\text{cf})}|. \quad (51)$$

This schematic comparison illustrates how an instantaneous centrifugal deformation may become asymmetric. The actual displacements are determined by the complete projected centrifugal force and the molecular force constant matrix. The resulting $\mathcal{J}$ -dependent geometry determines the physical centrifugal deformation, while the freely chosen s_{111} specifies the orientation of the contact frame used to represent this response. Such an instantaneous classical configuration does not imply symmetry breaking of a stationary rotational eigenstate.

Panels (c) and (d) represent the same semiclassical rotational response in two different contact gauges. In panel (c), the contact frame remains fixed at its equilibrium orientation, $s_{111} = 0$, and the deformation is represented in part by the nonzero quartic coefficient R_6 . In panel (d), the momentum-dependent transformation $\mathcal{J}' = \mathcal{R}(\mathcal{J}) \cdot \mathcal{J}$ reorients the contact frame through

$$\delta\vartheta_z = 2s_{111}\mathcal{J}_x\mathcal{J}_y, \quad (52)$$

and s_{111} may be chosen so that $R'_6 = 0$.

3.4. Dependence between R_5 and R_6 and Quartic Spectroscopic Invariant $C_{R_5R_6}$

In cylindrical representation, there are six unreduced constants, which include three off-diagonal, δ_J, R_5, R_6 . Among them, $\delta_J \equiv -T_{202}$ is unaffected by the unitary transformation $U = \exp(iS_3)$ and therefore cannot be excluded as a dependent one. Two different choices of S_3 can either eliminate $R_6 \equiv T_{004}$ (A -reduction, the most natural choice, as the associated operator produces $\Delta K = \pm 4$ terms beyond rigid rotor matrix structure) or $R_5 \equiv \frac{1}{2}T_{022}$ (S -reduction, the only remaining option for quasi-symmetric tops, for which $B_x \approx B_y$).

Since quartic unreduced spectroscopic constants have intrinsic linear dependency, it can be eliminated after the reduction. Because of the continuous gauge freedom generated by s_{111} , as discussed in Sec. 3.3, the transformed (reduced) spectroscopic constants R'_5 and R'_6 are mathematically locked together. Indeed, after the first unitary transformation reduced constants read:

$$R'_5 = R_5 - \frac{1}{2}s_{111}(2B_z - B_x - B_y), \quad (53)$$

$$R'_6 = R_6 + \frac{1}{4}s_{111}(B_x - B_y). \quad (54)$$

Solving the second equation for an arbitrary gauge parameter s_{111} yields:

$$s_{111} = 4 \left(\frac{R'_6 - R_6}{B_x - B_y} \right). \quad (55)$$

Now, after substituting this expression for s_{111} back into the first equation and rearranging these terms, we obtain the following quartic spectroscopic invariant:

$$R'_5 + 2R'_6 \left(\frac{2B_z - B_x - B_y}{B_x - B_y} \right) = R_5 + 2R_6 \left(\frac{2B_z - B_x - B_y}{B_x - B_y} \right) = C_{R_5 R_6}. \quad (56)$$

The ratio in parentheses is known in the literature as the asymmetry parameter σ [23] (should not be confused with Ray's asymmetry parameter $\varkappa = (2B - A - C)/(A - C)$) [52]:

$$\sigma = \frac{2B_z - B_x - B_y}{B_x - B_y}. \quad (57)$$

The right hand side in (56) with unreduced constants can be trivially evaluated from the molecular geometry and force field to provide a value of invariant $C_{R_5 R_6}$. Eq (56) also proves that, strictly speaking, it is meaningless to adjust both R_5 and R_6 to experimental data simultaneously using LSQ fitting, unless an additional constraint is introduced. One may also notice that in the case of SO_2F_2 the fraction in σ divides values of the same order of magnitude and hence produces a stable result, as a zero denominator may not correspond to an asymmetric rotor.

Using the $C_{R_5 R_6}$ invariant, we can see how the constants R_5 and R_6 are expressed through another one under Watson's standard reductions. For the A -reduction ($R'_6 = 0$), the remaining constant R'_5 absorbs the redundant parameter and is given by:

$$R'_5 = R_5 + 2R_6 \left(\frac{2B_z - B_x - B_y}{B_x - B_y} \right) = C_{R_5 R_6}. \quad (58)$$

This interesting result means that in the case of A -reduction the invariant $C_{R_5 R_6}$ simply equals R'_5 . In the alternative case of S -reduction ($R'_5 = 0$), R'_6 is expressed as:

$$R'_6 = R_6 + \frac{1}{2}R_5 \left(\frac{B_x - B_y}{2B_z - B_x - B_y} \right). \quad (59)$$

Now let us check the usefulness of these relations between R'_5 and R'_6 in the case of SO_2F_2 molecule. We will take the experimental effective constants A, B, C in S -reduction from Ref. [14] (microwave (MW) and by millimeter-wave (MMW) dataset from Table 1), and at this point we will ignore contributions to B_α from T_{022}, T_{004} that will be done below. In the I^r representation, the standard mapping of the principal rotational axes (a, b, c) to the Cartesian axes (x, y, z) is:

$$\begin{aligned} x \leftrightarrow b &\implies B_x = B = 0.169219657 \text{ cm}^{-1}, \\ y \leftrightarrow c &\implies B_y = C = 0.168685234 \text{ cm}^{-1}, \\ z \leftrightarrow a &\implies B_z = A = 0.171281068 \text{ cm}^{-1}. \end{aligned} \quad (60)$$

Armed with these values, we obtain stable and meaningful quantities despite of the singular character of reductions of SO_2F_2 :

$$\begin{aligned} \sigma &= \frac{2B_z - B_x - B_y}{B_x - B_y} = \frac{0.004657245}{0.000534423} \approx \mathbf{8.7145295}, \\ \sigma^{-1} &= \frac{B_x - B_y}{2B_z - B_x - B_y} = \frac{0.000534423}{0.004657245} \approx \mathbf{0.1147509}. \end{aligned} \quad (61)$$

Substituting the calculated ratio of σ to (58) and (59), we obtain

$$R'_5 = R_5 + 17.429R_6; \quad R'_6 = R_6 + 0.05737R_5. \quad (62)$$

Now it becomes evident why S -reduction is the preferred choice for SO_2F_2 molecule. If we keep $R'_6 \neq 0$, where R_5 and R_6 are of comparable absolute value, the second equation looks more numerically stable than one for $R'_5 \neq 0$, as any tiny numerical fluctuation or uncertainty in R_6 is multiplied by over 17 times when projected into R'_5 .

3.5. Consistency of Determining s_{111} from the Sextic Hamiltonian

The twice-transformed sextic component of the rotational Hamiltonian reads:

$$H'_{06} = H_{06} + \underbrace{i[S_3, H_{04}] - \frac{1}{2}[S_3, [S_3, H_{02}]]}_{\text{first order}} + \underbrace{i[S_5, H_{02}]}_{\text{second order}}. \quad (63)$$

It can be shown that P'_{006} , being a part of H'_{06} operator coefficients, after two contact transformations is given by the following expression:

$$P'_{006} = P_{006} + 2T_{004}s_{111} + B_{002}s_{111}^2 + \frac{1}{4}B_{002}(s_{311} - s_{131}). \quad (64)$$

Sarka *et al.* (2000) [13] proposed determining s_{111} from (64) by approximating $P'_{006} \approx 0$ while omitting both the S_5 -dependent contribution and the term quadratic in s_{111} . We therefore examine all contributions to P'_{006} in (64).

Indeed, the term $2T_{004}s_{111}$ appears after the first contact transformation from the commutator $i[S_3, H_{04}]$. The quadratic term, $B_{002}s_{111}^2$, also appears in the first order from $-\frac{1}{2}[S_3, [S_3, H_{02}]]$. The remaining term, proportional to $s_{311} - s_{131}$, originates from the third commutator $i[S_5, H_{02}]$. The transformation can be checked by substituting the A -reduction values of s_{111} , s_{311} , s_{131} into (64):

$$\begin{aligned} s_{111} &= -T_{004}/B_{002} = -\frac{1}{4}(T_{xx} + T_{yy} - 2T_{xy})/(B_x - B_y), \\ s_{311} &= -B_{002}^{-1}[2P_{006} + P_{204} - B_{002}^{-1}T_{004}(2T_{004} + T_{202})], \\ s_{131} &= B_{002}^{-1}[2P_{006} - P_{204} - B_{002}^{-1}T_{004}(2T_{004} - T_{202})]. \end{aligned} \quad (65)$$

It can be shown that (64) simplifies to the resulting expression identically collapses to zero:

$$P'_{006} = P_{006} - \frac{2T_{004}^2}{B_{002}} + \underbrace{\frac{T_{004}^2}{B_{002}} + \left(\frac{T_{004}^2}{B_{002}} - P_{006}\right)}_{\text{dropped}} \rightarrow 0. \quad (66)$$

where the underbrace marks the contributions omitted by Sarka *et al.* This identity shows that the omitted contributions cannot be discarded solely on order grounds; they are required for the exact cancellation obtained under the A -reduction conditions.

In Sarka's *et al.* work, another indirect fixing of s_{111} occurs after introducing the condition $h_1 = 0$. Finding a minimum of RMSD does not define s_{111} by itself, because RMSD is constant along the gauge orbit in the infinite order. The fixing happens due to cutting the contact transformations after the second order.

The transformed sextic coefficient can be expanded in series at the point $s_{111} = 0$:

$$h'_1(s_{111}) = h_1^{(0)} + h_1^{(1)}s_{111} + h_1^{(2)}s_{111}^2 + \dots; \quad h_1^{(n)} = \left. \frac{\partial^n h'_1}{\partial s_{111}^n} \right|_{s_{111}=0}. \quad (67)$$

546 If $h_1^{(1)} \neq 0$, the constraint $h_1' = 0$ selects the intersection of the gauge orbit with the constraint
 547 surface and the linear term is given by:

$$s_{111}^* = -h_1^{(0)}/h_1^{(1)}. \quad (68)$$

548 Since at sextic powers of J there are additional contact generators such as s_{311} , s_{131} and s_{113} ,
 549 $h_1' = 0$ fixes s_{111} uniquely only after fixation the other transformation parameters.

550 Now, we can make a number of conclusions. The proposed determination of s_{111} is not sup-
 551 ported by a consistent use of (64) for the following reasons:

552 1. First, in (64) both second ($2T_{004}s_{111}$) and third ($B_{002}s_{111}^2$) terms in the right side are
 553 generated by the S_3 transformation; both must therefore be retained. In this case, if the last,
 554 S_5 -dependent term is omitted, the condition $P'_{006} \approx 0$ yields a quadratic equation for s_{111} , and
 555 selection of an admissible root would require an additional criterion.

556 2. Second, the strict condition $P'_{006} = 0$ is distinctly associated with A -reduction (along with
 557 $T'_{004} = P'_{204} = P'_{024} = 0$), which was rejected by Sarka *et al.* as unsuitable because of potentially
 558 divergent series (s_{111} is larger than tolerance). Since with the A -reduction P'_{006} identically vanishes
 559 (*cf.* (66)), in this framework the approximate equation $P'_{006} \approx 0$ principally cannot provide an
 560 additional independent way of determination of s_{111} .

561 3. Third, the parameter s_{111} , being a part of $\hat{S}_3$, is responsible for modifying quartic con-
 562 stants by construction. Once s_{111} has been chosen, its effect is propagated into H'_{06} , while S_5 is
 563 defined by the sextic convention. If parameters s_{111} are to be defined using sextic constants, the
 564 corresponding operators should be included in the first perturbation, what would break down the
 565 entire perturbation scheme. In principle, choosing s_{111} through a sextic condition is possible as
 566 a gauge fixation (we can call it the ‘delayed’ reduction), but it artificially keeps both R_5 and R_6
 567 nonzero, which, however, does not increase the dimension of space of determinable parameters.

568 4. Finally, the suggestion to fix the S -reduction constant h_1 cannot be associated neither
 569 with A -, nor S -reduction, because h_1 originates from Φ_{402} sextic constant that does not disappear
 570 within any of two conventional reduction schemes. Imposing $h_1 = 0$ is therefore an additional
 571 constraint on that reduced model, rather than a new reduction by itself. If this constraint is
 572 used to select s_{111} and the parameters of S_5 , the resulting values are conditional on the imposed
 573 constraint.

574 If, indeed, the introduced constraint is $h_1 = 0$, which means introducing a new unconventional
 575 reduction, then the parameter s_{111} and quintic parameters s_5 should be obtained specifically using
 576 the corresponding equation. We can draw a conclusion that the choice of s_{111} is insufficiently
 577 supported theoretically and merely serves as choosing a suitable value, which is legitimate as a
 578 gauge fixation, but does not present a systematic solution.

579 3.6. Conditional Least Squares: Effect of the Constraint $h_1 = 0$ on Confidence Intervals

580 One of the essential reasons proving the validity of the 6-reduction was that the LSQ algorithm
 581 returned narrow standard deviations (σ_k) for the unreduced constants R_5 and R_6 [12, 13]. However,
 582 the narrow standard deviations were obtained after introducing an auxiliary conditional constraint,
 583 $h_1 = 0$. To understand the influence of an imposed constraint on the confidence intervals of the
 584 fitted parameters, it is convenient to consider the following geometrical interpretation.

585 Let us take a hybrid sextic effective Hamiltonian, composed of $n = 16$ parameters, including
 586 rotational constants, unreduced quartic parameters (proportional to T_{klm}) and seven S -reduced
 587 sextic parameters. Let $\mathbf{p} = (A, B, C, T_{400}, \dots, T_{004}, H_1, \dots, h_3)^\top$ denote the vector of three rota-
 588 tional, six quartic, and seven sextic S -reduced parameters of the effective Hamiltonian. A varia-
 589 tion of the contact transformation parameter s_{111} moves these parameters along a one-dimensional
 590 curvilinear gauge orbit, whose local tangent is the gauge vector $\mathbf{g} = \partial\mathbf{p}/\partial s_{111}$. The dependence of

transition frequencies $\nu(\mathbf{p})$ on variation of parameters $\mathbf{p}$ is described by the Jacobian $\mathbb{J} = \partial\nu/\partial\mathbf{p}$, so that the following local null-space condition is obeyed:

$$\frac{\partial\nu(\mathbf{p})}{\partial s_{111}} = \mathbb{J} \cdot \mathbf{g} \simeq 0, \quad (69)$$

as displacements along the gauge orbit leave the spectrum unchanged within the considered truncated Hamiltonian. The least squares normal matrix $\mathbb{G}$ satisfies the corresponding condition:

$$\mathbb{G} = \mathbb{J}^\top W \mathbb{J}, \quad \mathbb{G} \cdot \mathbf{g} = \mathbb{J}^\top W (\mathbb{J} \cdot \mathbf{g}) \simeq 0. \quad (70)$$

Geometrically, the joint confidence region therefore resembles a narrow tunnel directed along $\mathbf{g}$, the cross-section of which represents the finite uncertainty of the spectroscopically estimable parameter combinations, perpendicular to the gauge direction. Within the local linearized description, an exact gauge freedom makes this tunnel unbounded along $\mathbf{g}$, and every parameter having a nonzero component along $\mathbf{g}$, including R_5 and R_6 , becomes individually non-estimable.

If the gauge freedom is the only singular direction, matrix $\mathbb{G}$ can be expressed in an equivalent form via its eigenvalues and eigenvectors:

$$\mathbb{G} = \sum_{i=1}^{n-1} \lambda_i \mathbf{u}_i \mathbf{u}_i^\top + \lambda_g \hat{\mathbf{g}} \hat{\mathbf{g}}^\top, \quad \mathbf{u}_i^\top \mathbf{u}_j = \delta_{ij}, \quad \mathbf{u}_i^\top \hat{\mathbf{g}} = 0, \quad \lambda_g \simeq 0. \quad (71)$$

where $\hat{\mathbf{g}}$ denotes the normalized gauge vector. The formal standard uncertainty of each parameter p_k is a projection of the complete confidence tunnel onto the p_k coordinate axis:

$$\sigma_k^2 \simeq s_\nu^2 \left(\sum_{i=1}^{n-1} \frac{u_{ik}^2}{\lambda_i} + \frac{\hat{g}_k^2}{\lambda_g} \right), \quad s_\nu^2 = (N_{\text{obs}} - \text{rank}(\mathbb{J}))^{-1} \sum_{j=1}^{N_{\text{obs}}} w_j (\nu_j^{\text{obs}} - \nu_j^{\text{calc}})^2. \quad (72)$$

It follows from this equation that, for every parameter having $\hat{g}_k \neq 0$, the formal standard uncertainty becomes unbounded as the gauge eigenvalue $\lambda_g \rightarrow 0^+$. For a truncated numerical model, λ_g may be small, producing large but finite uncertainties.

The imposed condition $h_1(\mathbf{p}) = 0$ removes one independent parameter variation and defines a hypersurface in the same parameter space. The vector $\mathbf{n} = \nabla_{\mathbf{p}} h_1$ is perpendicular to the constraint surface, whereas a variation of parameters satisfying $\delta h_1 = \mathbf{n}^\top \delta \mathbf{p} = 0$ lies within this surface. The variation of h_1 along the gauge orbit is then given by:

$$\frac{dh_1}{ds_{111}} = \mathbf{n}^\top \mathbf{g}. \quad (73)$$

This relation determines whether the gauge orbit crosses the surface or runs tangentially along it. If the gauge orbit pierces the surface $h_1 = 0$, this constraint locally selects one point on the otherwise isospectral orbit.

Introducing an $[n \times (n-1)]$ matrix $\mathbf{Z}$, whose columns span the tangent space of $h_1 = 0$ and therefore satisfy $\mathbf{n}^\top \mathbf{Z} = 0$, we can describe the allowed variations of original coordinates via a reduced set of $n-1$ independent coordinates $\delta \mathbf{q}$:

$$\delta \mathbf{p} = \mathbf{Z} \delta \mathbf{q}. \quad (74)$$

Under the condition $\mathbf{n}^\top \mathbf{g} \neq 0$, the matrix $\mathbf{Z}$ restricts the least-squares problem to the local tangent space of $h_1 = 0$, which does not contain the complete gauge direction.

619 The matrix $\mathbf{Z}$ is not unique; any set of $n - 1$ linearly independent columns satisfying $\mathbf{n}^\top \mathbf{Z} =$
620 0 provides a valid basis of the tangent space. Its structure becomes explicit if one parameter
621 variation, denoted by δp_1 , is eliminated using the constraint:

$$n_1 \delta p_1 + \mathbf{n}_v^\top \delta \mathbf{p}_v = 0, \quad \delta p_1 = -\frac{\mathbf{n}_v^\top}{n_1} \delta \mathbf{p}_v. \quad (75)$$

622 Choosing $\delta \mathbf{q} = \delta \mathbf{p}_v$, the condition $\delta \mathbf{p} = \mathbf{Z} \delta \mathbf{q}$ takes the explicit form

$$\delta \mathbf{p} = \begin{pmatrix} -\mathbf{n}_v^\top / n_1 \\ \mathbf{I}_{n-1} \end{pmatrix} \delta \mathbf{q}, \quad \mathbf{n}^\top \mathbf{Z} = 0. \quad (76)$$

623 Each column of $\mathbf{Z}$ therefore describes the complete change of the original parameter vector pro-
624 duced by varying one independent coordinate q_k while preserving $h_1 = 0$. If h_1 is explicitly one
625 component of $\mathbf{p}$, then $\mathbf{n}$ is simply the unit vector along h_1 , and $\mathbf{Z}$ is an identity matrix with the
626 h_1 row replaced by zero.

627 Having expressed all allowed parameter variations in terms of $\delta \mathbf{q}$, we must also express the
628 least-squares quadratic form in these reduced coordinates. Substitution of $\delta \mathbf{p} = \mathbf{Z} \delta \mathbf{q}$ gives

$$\delta \mathbf{p}^\top \mathbb{G} \delta \mathbf{p} = \delta \mathbf{q}^\top (\mathbf{Z}^\top \mathbb{G} \mathbf{Z}) \delta \mathbf{q}. \quad (77)$$

629 which identifies $\mathbb{G}_q = \mathbf{Z}^\top \mathbb{G} \mathbf{Z}$ as the normal matrix for the constrained fit. Its inverse gives the
630 covariance in the reduced coordinates, which is subsequently transformed back to the original
631 parameter space by $\mathbf{Z}$.

632 The constrained covariance matrix is then given by:

$$\mathbb{G}_q = \mathbf{Z}^\top \mathbb{G} \mathbf{Z}, \quad \mathbb{C}_q = s_v^2 \mathbb{G}_q^{-1}, \quad \mathbb{C}_{h_1=0} = \mathbf{Z} \mathbb{C}_q \mathbf{Z}^\top = s_v^2 \mathbf{Z} (\mathbf{Z}^\top \mathbb{G} \mathbf{Z})^{-1} \mathbf{Z}^\top. \quad (78)$$

633 For a locally linear weighted least-squares problem, (78) is the standard tangent-space form of the
634 covariance matrix under a differentiable equality constraint [53]. The multiplication by $\mathbf{Z}$ restricts
635 the fit to variations tangent to $h_1 = 0$ and removes the gauge direction before the reduced normal
636 matrix is inverted.

637 One may imagine the constraint surface as a thin sheet cutting across the long confidence
638 tunnel. In appropriately scaled parameter coordinates, the angle θ between the gauge vector and
639 the tangent surface is defined by

$$\sin \theta = \frac{|\mathbf{n}^\top \mathbf{g}|}{\|\mathbf{n}\| \|\mathbf{g}\|}. \quad (79)$$

640 When the sheet is nearly perpendicular to the tunnel ($\theta \approx 90^\circ$), the constrained least-squares
641 calculation consequently produces narrow formal confidence intervals. In contrast, if θ approaches
642 zero, the gauge direction becomes tangent to the constraint surface and the singularity is not
643 removed. Figure 2 illustrates a model three-parameter system, where introduction of a constraint
644 transforms infinite confidence intervals into finite ones in a lower dimension.

645 This effect can also be seen by decomposing a local parameter displacement and applying the
646 constraint, where $\delta \mathbf{p}_\perp$ is perpendicular to $\mathbf{g}$ in the scaled parameter coordinates:

$$\delta \mathbf{p} = \mathbf{g} \delta s_{111} + \delta \mathbf{p}_\perp, \quad \delta s_{111} = -\frac{\mathbf{n}^\top \delta \mathbf{p}_\perp}{\mathbf{n}^\top \mathbf{g}}. \quad (80)$$

647 The gauge coordinate δs is therefore no longer determined by the spectrum, but is algebraically
648 fixed by the constraint and by the remaining parameter variations. However, the spectrum has

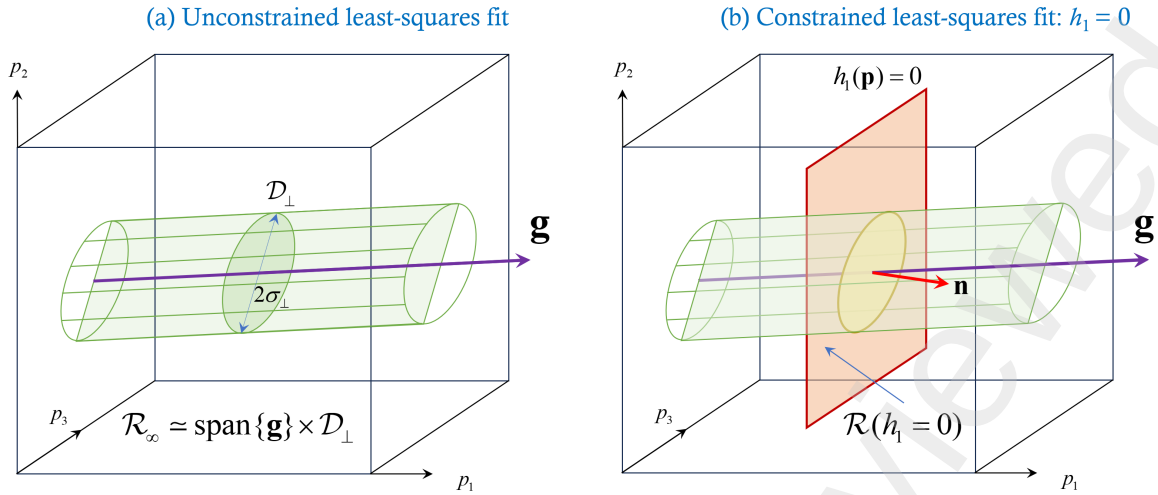


Figure 2:

An unbounded joint confidence tunnel $\mathcal{R}_\infty$ for the unconstrained LSQ fit (Panel (a)) becomes a finite region $\mathcal{R}_{h_1=0} = \mathcal{R}_\infty \cap \{\mathbf{p}: h_1(\mathbf{p}) = 0\}$, shown on Panel (b) by imposing the constraint $h_1 = 0$. Because $\mathbb{J} \cdot \mathbf{g} \simeq 0$ and eigenvalue $\lambda_g \rightarrow 0$, its confidence region is unbounded along $\mathbf{g}$ with a finite cross-section $\mathcal{D}_\perp$. Panel (b) shows the constraint surface with normal vector $\mathbf{n} = \nabla_{\mathbf{p}} h_1$. When $dh_1/ds_{111} = \mathbf{n}^\top \mathbf{g} \neq 0$, the surface cuts across the confidence tunnel and excludes its unlimited direction. The remaining conditional covariance is $\mathbb{C}_{h_1=0} = s_\nu^2 \mathbf{Z}(\mathbf{Z}^\top \mathbb{G} \mathbf{Z})^{-1} \mathbf{Z}^\top$, where columns of matrix $\mathbf{Z}$ span the tangent space, $\mathbf{n}^\top \mathbf{Z} = 0$.

supplied no additional information along the gauge direction: the unbounded part of the confidence region has merely been excluded by the imposed condition.

The resulting uncertainties are therefore conditional quantities,

$$\sigma_k = \sigma_k(\text{spectroscopic data} \mid h_1 = 0), \quad (81)$$

and not uncertainties determined by the spectrum alone. Their small values demonstrate the numerical stability of the fit within the selected reduction convention, but they cannot establish the independent physical determinability of all six quartic constants. Thus, narrow confidence intervals obtained after imposing $h_1 = 0$ may be deceptively interpreted as evidence of physical meaningfulness, whereas the spectroscopically meaningful quantities are gauge-invariant combinations, such as $C_{R_5 R_6}$, that remain unchanged along the gauge orbit.

3.7. Subsequent Treatments of the Indeterminacy Problem – ITO Approach

The analysis of Sarka *et al.* was followed by several studies of SO_2F_2 [9, 14–21, 54] and its isotopic species $\text{S}^{18}\text{O}_2\text{F}_2$ [12]. Many of which employed an irreducible tensor operator (ITO) formalism adapted to the $\text{O}(3) \supset T_d \supset C_{2v}$ group chain [9, 15, 17, 18, 20, 21, 54]. These works substantially extended measured frequency ranges, and established useful relations between the tensorial and Watson-type Hamiltonians. Nevertheless, the distinction between numerical determinability within a constrained model and genuine independence of spectroscopic constants remained unresolved.

In the continuation of the original work, Bürger *et al.* [14] quoted recent Demaison’s unpublished SO_2F_2 MW data for higher J, K_a and repeated the conclusion that ‘6-reduction is “clearly superior” to S -reduction in that domain of quantum numbers.’ Their actual analysis, however, employed Watson’s S -reduction, because J, K_a values for the ν_8 excited state ‘are not high enough to demand 6-reduction’. Subsequent Watson-type analyses of the (ν_3, ν_7, ν_9) bending triad and the ν_2 state retained the S -reduction [16, 19]. Bürger *et al.* [16] cautioned that some coefficients

672 multiplying higher powers of J were effective corrections compensating deficiencies of the adopted
 673 Coriolis model, while Merke *et al.* [19] fixed three sextic coefficients to their ground-state values.
 674 These two works did not discuss the validity of the 6-reduction.

675 Rotger *et al.* (2002) [15] subsequently introduced the ITO formalism for quasi-spherical XY_2Z_2
 676 molecules. Citing Sarka's *et al.* work, the authors acknowledged that the analysis of available
 677 SO_2F_2 spectra by conventional methods does not seem to be totally satisfying. Their preliminary
 678 ground-state analysis used 806 transitions extending up to $J = 99$, but one of the six quartic ten-
 679 sorial parameters ($\Omega(K, n\Gamma, \tilde{\Gamma}) = 4(4, 0 F_2, a_1) = t_9$) was fixed. In the subsequent more systematic
 680 comparison of the tensorial and Watson Hamiltonians [17], the relations between the parameters of
 681 both tensorial and Watson's models were established. In this work, the 1071 assigned transitions
 682 were fitted to the Watsonian using I^r representation in the S -reduction and quartic 6-reduction
 683 with additional varied all seven sextic constants in S -reduction, removing the constraint $h_1 = 0$.
 684 Applying the tensorial Hamiltonian model, the same tensorial parameter t_9 was fixed at the spe-
 685 cific value, obtained by optimizing the residual as a function of t_9 . The alternative condition
 686 (' $t_9 = 0$ Reduction'), corresponding to $T_{022} = 2R_5 = 0$ (same as in Watson's S -reduction) was also
 687 considered because of the 'ambiguity problem.' Rotger *et al.* (2003) [17] confirmed the conclusion
 688 of Sarka's *et al.* work that, in the particular case of SO_2F_2 , it is preferable to use an unreduced
 689 Hamiltonian (at least up to quartic terms). In this work it was noted that the rotational constants
 690 have a smaller standard deviation in the 6-reduction and that the sextic constants are significantly
 691 different in the S - and 6-reductions. The subsequent work by Rotger *et al.* (2006) [18] confirmed
 692 that the standard Watsonian is not satisfactory and in particular the A - and S -reductions in the
 693 ground state analysis, while 'the best results are obtained using the so-called 6-reduction'. In
 694 confirmation of this view, the authors cite Mekhtiev and Hougen's (2000) work [55], where the
 695 authors showed that, when a conventional quartic contact transformation generates large sextic
 696 and higher-power tails, a partially reduced Hamiltonian retaining all six quartic coefficients may
 697 provide a numerically superior fit. Their delayed-reduction example fitted six quartic and nine
 698 sextic coefficients with an acceptable condition number, whereas introduction of the tenth sextic
 699 coefficient restored severe ill-conditioning. This result does not establish a sixth gauge-invariant
 700 quartic observable. Rather, it demonstrates that the reduction constraint can be displaced from
 701 the quartic to the sextic power, so that the resulting narrow parameter uncertainties remain
 702 conditional on the imposed sextic constraint.

703 The detailed correspondence between the tensorial and Watson Hamiltonians, derived system-
 704 atically by Boudon *et al.* (2006) [9] provides an important bridge between the two formalisms.
 705 In particular, for both A - and S -reduction the relationships between eliminated cylindrical con-
 706 stants T'_{klm}, P'_{klm} and tensorial constants t'_n were derived. The 6-reduction was described as the
 707 S -reduction without imposing $T'_{022} = 0$.

708 In a later work by Margules *et al.* (2009) [12] on species $S^{18}O_2F_2$, six quartic coefficients were
 709 again reported as determinable 'due to the near-sphericity of the molecule', and the agreement
 710 with values calculated from an *ab initio* force field was taken as support for the analysis, which
 711 is 'another indication of the absence of problems in the fits'. Small confidence intervals ($|p_k| >$
 712 $60\sigma(p_k)$) confirmed 'that the high-correlations between the parameters are not harmful'.

713 Later C_{2v} Top Data System analyses extended the ITO treatment to new line-position and
 714 intensity data for ν_2 and the bending triad [20, 21, 54]. These fits retained selected tensorial
 715 coefficients at zero or at earlier values, and the authors correctly noted that such constraints
 716 are not identical to Watson's A - or S -reduction conditions. They validate the computational
 717 effectiveness of the ITO approach, but do not provide a Jacobian-rank test of six freely varied
 718 ground-state quartic coefficients.

719 More generally, an invertible transformation between Watson and ITO coefficient vectors can-
 720 not change the rank of the spectroscopic Jacobian. Consequently, the ITO technique neither

creates nor removes the quartic gauge freedom. Its considerable advantages for symmetry adaptation, polyad construction, intensity calculations, and spectrum simulation remain fully valid, but the fitted tensorial coefficients must still be interpreted in the view of contact transformations.

We can draw a conclusion that aforementioned post-2000 Sarka's work analyses therefore supplied extensive experimental and computational confirmation of the effectiveness of nonstandard parametrizations, but they still did not demonstrate that the sixth quartic coefficient is an independent gauge-invariant observable. Taking these works as typical examples, we can see that confidence intervals and the residual RMSD universally served as ultimate criterions of validity a certain effective Hamiltonian with implicitly assumed correct correspondence to the underlying physical models.

4. Calculations

In this Section, we will first provide the key theoretical details of the essential work of Dreizler and Dendl (1965) [25] (hereafter referred to as DD1965), which is, to our knowledge, the earliest systematic study of the linear dependence of six quartic centrifugal distortion constants $A_1 \dots A_6$ of an asymmetric top, earlier proposed by Kivelson and Wilson [26]. We will further apply their technique for investigating the manifestation of effects, observed in DD1965, using ground state MW data of SO_2F_2 [13]. Nowadays, Kivelson and Wilson's approach to modeling quartic rotational energy W_q is no longer in use, because it is more complex and its aim was avoiding matrix diagonalizations. Although this archive method, relying on expectation values $\langle P_z^{2n} \rangle$ and parameters $A_1 - A_6$, is no longer in use in modern computational molecular spectroscopy, the underlying ideas of DD1965 work can be successfully applied to analyzing more customary sextic centrifugal distortion parameters in cylindrical representation T_{klm} , which is also accomplished.

4.1. Kivelson and Wilson's Parameters $A_1 - A_6$ and Their Linear Dependence

At first, DD1965 performed a simultaneous least-squares (LSQ) fit of three rigid-rotor rotational constants A, B, C and Kivelson and Wilson's quartic centrifugal distortion constants $A_1 \dots A_6$ [26] to microwave (MW) experimental data of the $(\text{CH}_3)_2\text{SO}$ and $(\text{CD}_3)_2\text{SO}$ (dimethyl sulfoxide, or DMSO) isotopologues. In their notation, DD1965 adopted the standard III_r representation, where the principal axes x, y, z correspond to the rotational constants A, B, C , respectively.

Upon discovering a strong correlation between nine fitted parameters (A, B, C plus six $A_1 \dots A_6$), which manifested as poor confidence intervals, Dreizler and Dendl introduced another strategy. They varied the asymmetry parameter κ over a wide range near its optimal value κ_0 . For each fixed value of κ , they adjusted two rotational parameters $p_+ = (A + C)/2$ and $p_- = (A - C)/2$, along with six quartic constants $A_1 \dots A_6$. This construction decreased the number of LSQ fitted parameters to eight. When κ is constrained, the values of W_r/p_- , $\langle P_z^2 \rangle$, and $\langle P_z^4 \rangle$ become constant for every transition, which reduces the problem to a linear, well-defined system of equations.

Dreizler and Dendl presented the results of their calculations graphically, plotting three essential functional dependencies: $A_j(\kappa)$, the RMS deviation $\sigma(\kappa)$, and the centrifugal distortion contribution to rotational energy $\Delta\nu_c(\kappa) = E(J^2, J^4) - E(J^2)$. This form of presentation demonstrated that at the optimum point ($\kappa_0 = 0.9103$ for $(\text{CH}_3)_2\text{SO}$ and $\kappa_0 = 0.5129$ for $(\text{CD}_3)_2\text{SO}$), the contribution of the centrifugal distortion energy $\Delta\nu_c(\kappa)$ is minimized, while the functions $A_k(\kappa)$ nearly converge at a near-intersection point at κ_0 . At this point, they gain their most physically meaningful, near-minimum values. The functions $A_k(\kappa)$ cluster very close to zero on ordinate, because actual centrifugal distortion is a minor physical correction compared to the rigid-rotor rotational energy. At the same time, the RMS fitting error function $\sigma(\kappa)$ forms a curve with a very shallow, nearly flat valley. Since this early observation by Dreizler and Dendl is of central importance to the present work, let us reproduce the details of their theory and subsequent calculations.

768 The quartic centrifugal distortion energy in the Kivelson and Wilson formalism is given by
 769 the following expression, which depends on the quantum number J and the expectation values,
 770 which, according to the Hellmann-Feynman theorem [56], are given by $\partial^n W_r / \partial \alpha^n = \langle P_\alpha^{2n} \rangle$ [57]:

$$W_q = W_r + A_1 W_r^2 + A_2 W_r J(J+1) + A_3 J^2(J+1)^2 + A_4 J(J+1) \langle P_z^2 \rangle + A_5 \langle P_z^4 \rangle + A_6 W_r \langle P_z^2 \rangle, \quad (82)$$

771 where $W_r = B_x P_x^2 + B_y P_y^2 + B_z P_z^2 = \alpha P_z^2 + \beta P_x^2 + \gamma P_y^2$ is an asymmetric rigid-rotor energy, and
 772 P_λ ($\lambda = x, y, z$) represents the λ -component of the angular momentum operator, distinguishing the
 773 operator from the rotational quantum number J , for which $\langle P_\alpha^2 \rangle + \langle P_\beta^2 \rangle + \langle P_\gamma^2 \rangle = J(J+1)$. The
 774 expectation values $\langle P_z^2 \rangle, \langle P_z^4 \rangle$ relate to a given state. Eq. (82) is a convenient form for expressing
 775 the rotational energy because the unperturbed energy of an asymmetric rigid rotor is typically
 776 written as:

$$W_r = \frac{A+C}{2} J(J+1) + \frac{A-C}{2} E(\kappa) = p_+ J(J+1) + p_- E(\kappa), \quad (83)$$

777 where $E(\kappa)$ is the eigenvalue of the reduced Hamiltonian matrix for a specific κ . By fixing κ ,
 778 the rotational constants A, B, C become linked together. For given values of p_+, p_- , and κ , these
 779 constants can be reconstructed as: $A = p_+ + p_-$, $B = p_+ + \kappa p_-$, and $C = p_+ - p_-$.

780 The quartic constants $A_1 \dots A_6$ can be expressed in terms of the more customary operator-
 781 dependent cylindrical parameters used in Sarka *et al.*'s formulation. First, the rigid-rotor energy
 782 operator is defined as:

$$W_r = B_{200} \mathbf{J}^2 + B_{020} J_z^2 + B_{002} (J_+^2 + J_-^2). \quad (84)$$

783 Substituting W_r from (84) into the Kivelson and Wilson expression (82), we obtain:

$$W_c = (B_{200} \mathbf{J}^2 + B_{020} J_z^2 + \dots) + A_1 (B_{200} \mathbf{J}^2 + B_{020} J_z^2 + \dots)^2 + A_2 (B_{200} \mathbf{J}^2 + \dots) \mathbf{J}^2 + \dots \quad (85)$$

784 After expanding the expressions and matching the coefficients of identical operators, we obtain the
 785 relations for the constants $A_1 \dots A_6$ in terms of the quadratic and quartic cylindrical constants:

$$\begin{aligned} A_1 &= T_{004} / B_{002}^2, \\ A_2 &= T_{202} / B_{002} - 2T_{004} B_{200} / B_{002}^2, \\ A_3 &= T_{400} - T_{202} B_{200} / B_{002} + T_{004} B_{200}^2 / B_{002}^2, \\ A_4 &= T_{220} - (T_{202} B_{020} + T_{022} B_{200}) / B_{002} + 2T_{004} B_{200} B_{020} / B_{002}^2, \\ A_5 &= T_{040} - T_{022} B_{020} / B_{002} + T_{004} B_{020}^2 / B_{002}^2, \\ A_6 &= T_{022} / B_{002} - 2T_{004} B_{020} / B_{002}^2. \end{aligned} \quad (86)$$

786 Watson [3] performed a thorough analysis of Dreizler and Dendl's results, though the funda-
 787 mental proof of the indeterminacy of the six quartic centrifugal distortion constants had been
 788 established in his earlier work [1]. He identified the following linear identity between the expecta-
 789 tion values ($\alpha = C, \beta = A, \gamma = B$):

$$\begin{aligned} 0 &= [W_r - \beta J(J+1)] [W_r - \gamma J(J+1)] - 2[(2\alpha - \beta - \gamma)W_r \\ &\quad - (\alpha\beta + \alpha\gamma - 2\beta\gamma)J(J+1)] \langle P_z^2 \rangle + 3(\alpha - \beta)(\alpha - \gamma) \langle P_z^4 \rangle \\ &\quad + (\alpha - \gamma)(\beta - \gamma) \langle P_x^2 \rangle - (\alpha - \beta)(\beta - \gamma) \langle P_y^2 \rangle - (\beta - \gamma)^2 \langle P_z^2 \rangle. \end{aligned} \quad (87)$$

790 An arbitrary multiple ξ of the right-hand side of (87) can be added to the right-hand side of (82).
 791 In this manner, Watson demonstrated that the parameters $A_1 \dots A_6$ can be modified by adding a

792 scaled null vector $c \cdot \xi$ in a consistent manner, leaving the observable transition energies intact:

$$\begin{aligned}
 A'_1 &= A_1 + \xi, \\
 A'_2 &= A_2 - (\beta + \gamma) \xi, \\
 A'_3 &= A_3 + \beta\gamma \xi, \\
 A'_4 &= A_4 + 2(\alpha\beta + \alpha\gamma - 2\beta\gamma) \xi, \\
 A'_5 &= A_5 + 3(\alpha - \beta)(\alpha - \gamma) \xi, \\
 A'_6 &= A_6 - 2(2\alpha - \beta - \gamma) \xi.
 \end{aligned} \tag{88}$$

793 These analytical expressions successfully reproduce the parametric curves obtained by DD1965.
 794 Indeed, the point $\xi = 0$ corresponds to the near-intersection of functions $A_k(\kappa)$, where $\sum_j A_j^2 =$
 795 min, while varying ξ across a small interval (e.g., from -0.001 to $+0.001$) generates nearly linear
 796 dependence on parameter ξ .

797 The underlying physical reason why this intersection occurs precisely at κ_0 now becomes ap-
 798 parent. As the predefined parameter κ is swept along the horizontal axis, the model tests different
 799 contributions for the rigid-rotor frame to the observed spectrum. When $\kappa \neq \kappa_0$, the rigid-rotor
 800 geometry is incorrect. To compensate for this geometric error, the A_i parameters are artificially
 801 scaled, causing them to deviate sharply along the vertical axis of the parameter plot. In contrast,
 802 when $\kappa = \kappa_0$, the geometry corresponds the actual molecular structure and the calculated con-
 803 tributions of the rigid rotor best match the experimental data. Consequently, the A_i parameters
 804 relax and drop down to their small, physically realistic values, forming a coordinated convergence
 805 point on the graph.

806 4.2. Linear Dependence of $A_1 - A_6$ and Quartic Constants T_{klm} for SO_2F_2

807 To study functional dependence of the $A_1 - A_6$ constants, T_{klm} , RMSD (σ) and $\Delta\nu_{J^4}$ on the
 808 asymmetry parameter κ , the ground state rotational levels of SO_2F_2 were used. Since experimental
 809 MW data in Ref. [13] are provided as transitions, the required rotational levels were derived from
 810 the empirical constants of the S -reduced Watson Hamiltonian, taken from Table N^o1 in Rotger
 811 *et al.* (2003) publication [17], where the authors claim that their effective Hamiltonian describes
 812 ground state levels with $\text{RMSD} = 3.4 \times 10^{-6} \text{cm}^{-1}$. The set of generated rotational levels was limited
 813 by setting $J_{\max} = (K_a)_{\max} = 45$, because further increase in $J_{\max}$ and $(K_a)_{\max}$ lead to significant
 814 deterioration of reproducing observed energy terms within the quartic effective Hamiltonian model.

Next, we repeated the numerical experiment of Dreizler and Dendl (1965) [25] on the SO_2F_2
 molecule. Substituting rotational constants from Table N^o1, taken from Ref. [13], $A = 5134.87727$,
 $B = 5073.06737$, and $C = 5057.06518$ MHz, to Ray's asymmetry parameter (κ), we obtain:

$$\kappa = \frac{2B - A - C}{A - C}; \quad \kappa_0 = \frac{-45.80771}{77.81209} \approx -0.5887,$$

815 which is in accord with their reported value of -0.589 .

816 Results of the full-scale modeling of functional dependence of fitted parameters $A_1 \dots A_6$ on
 817 varied κ near its experimental ground state equilibrium value κ_0 are presented in Figure 3. The left
 818 part of the Figure shows that near κ_0 values of $A_1 \dots A_6$ cluster near zero. With a more significant
 819 change of $\kappa < 0.65$ the functional dependency becomes slightly curvilinear, reflecting an existence
 820 of higher-order terms ($J^n, n > 4$). Coordinated, near linear changes of parameters $A_1 \dots A_6$,
 821 obtained by fitting the model to the experimental data by systematic alteration of κ , firmly prove
 822 an existence of redundancy in the model, containing three rotational and six quartic parameters.
 823 Examination of the right part of Figure 3 reveals a shallow, spurious minimum in the residual fit
 824 error. It can be seen that in a wide range of varied parameter, $-0.60 < \kappa < -0.55$, the residual

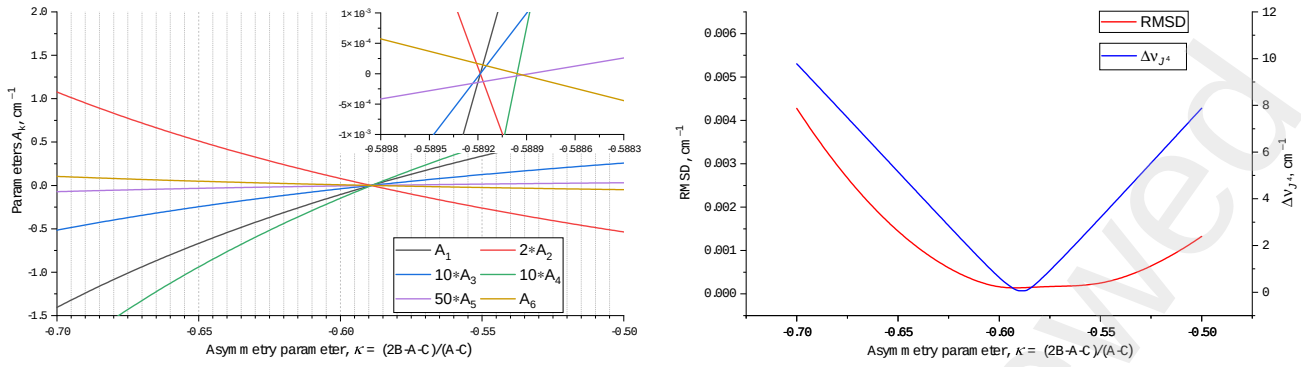


Figure 3: (Left Part) Fitted values of Kivelson and Wilson's scaled quartic parameters $A_1 \dots A_6$ as functions of pre-defined values of κ . The magnified part of functional dependencies demonstrates that at the point κ_0 constants take small and physically realistic values of the order 10^{-7} cm^{-1} .

(Right Part) Functional dependence of fitting RMSD (σ) as a function of κ . It can be seen that in the range of $\kappa = -0.60 \dots -0.55$ RMSD remains nearly constant, while constants $A_1 \dots A_6$ manifest variation in wide ranges. The contribution of purely quartic terms $\Delta\nu_{J^4}$, calculated as a difference $E(J^2, J^4) - E(J^2)$, show a narrow curvilinear minimum near κ_0 and practically linear dependence on κ outside this area.

fitting error remains nearly constant, which serves as an alternative proof of the redundancy in applied theoretical model. We also added the functional dependence of the contribution of quartic terms to reproduction of observed data $\Delta\nu_{J^4} = E(J^2, J^4) - E(J^2)$, to match Dreizler and Dendl calculation. It can be seen in the right part of Figure 3 that $\Delta\nu_{J^4}$ forms a smooth minimum near κ_0 an linear dependence for larger deviations of κ .

Our next numerical experiment concerned the form of functional dependency of more customary quartic centrifugal distortion parameters T_{klm} , $k + l + m = 4$ in cylindrical representation. These parameters are trivially connected to Nielsen's constants $D_J, D_{JK}, D_k, \delta_J, R_5, R_6$. Unlike parameters $A_1 \dots A_6$, which have a significant contribution from rigid-rotor energy by their construction, parameters T_{klm} have much smaller equilibrium values. The results of our numerical experiment, that fully repeats the logic of DD1965 calculation, are presented in Figure 4. A visual inspection of this Figure essentially repeat the same conclusions as it was done above on the basis of using parameters $A_1 \dots A_6$, except the fact that true values of T_{klm} are much smaller.

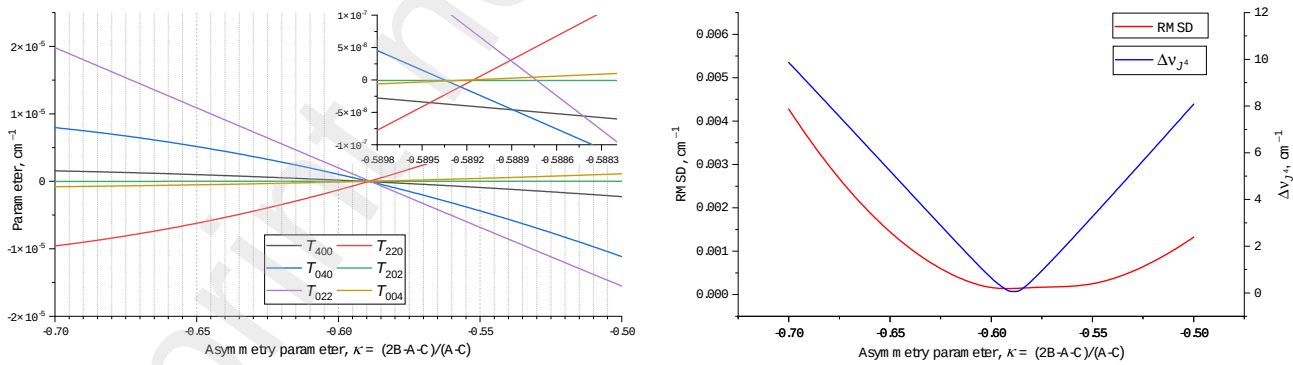


Figure 4: (Left Part) Fitted values of cylindrical quartic parameters $T_{400} \dots T_{004}$ as functions of pre-defined values of κ . The magnified part of functional dependencies demonstrate that at the point κ_0 constants take small and physically realistic values of the order 10^{-4} cm^{-1} .

(Right Part) Functional dependence of fitting RMSD (σ) as a function of κ . The contribution of purely quartic terms $\Delta\nu_{J^4} = E(J^2, J^4) - E(J^2)$, show a narrow curvilinear minimum near κ_0 and practically linear dependence on κ outside this area.

838 4.3. Interconnection between R_5 and R_6 and Invariant $C_{R_5R_6}$

839 We have seen above in Sec. 3.4 that the following first-order invariant function of R_5, R_6 and
840 rigid-rotor parameter $\sigma = (2B_z - B_x - B_y)/(B_x - B_y)$ is predicted theoretically:

$$R'_5 + 2R'_6 \left(\frac{2B_z - B_x - B_y}{B_x - B_y} \right) = R_5 + 2R_6 \left(\frac{2B_z - B_x - B_y}{B_x - B_y} \right) = C_{R_5R_6}. \quad (89)$$

841 An experimental verification of validity and accuracy of existence of this invariant constant $C_{R_5R_6}$
842 can unequivocally prove that within the so-called 6-reduction two parameters have first-order
843 linear relationship and therefore this model has one redundant parameter.

844 We conducted the following numerical experiment. We had chosen the hybrid effective Hamil-
845 tonian, containing A, B, C , six Nielsen's constants $D_J, D_{JK}, D_K, \delta_J, R_5, R_6$, supplemented by all
846 seven S -reduced sextic constants, without the constraint $h_1 = 0$. This form of the Hamiltonian
847 ensures a rigorous test of existence of linear dependence in six quartic constants and avoids hidden
848 influence of this constraint on the result. In capacity of the experimental data, we took the MW
849 assigned ground state transitions from Rotger *et al.* (2003) work [17] with limitations $J'_{\max} \leq 85$
850 and $K'_{a,\max} \leq 50$.

851 We performed a series of 40 fits, each time fixing the value of R_6 in wide range between
852 $0.0 - 1.0 \times 10^{-9} \text{ cm}^{-1}$, covering the double of our *ab initio* quantum mechanical value of $R_6^{\text{QC}} =$
853 4.4972×10^{-8} . In addition, the minimum RMSD was found by performing a separate fit in which all
854 constants, including R_6 , were varied simultaneously. For calculation of σ we used fitted constants
855 A, B, C , because their difference with corresponding ground state values of B_x, B_y, B_z is negligible
856 ($< 0.01\%$) for verifying the quartic invariant. The results are presented on Figure 5. It can be
857 seen that the combination $R_5 + 2\sigma R_6 = C_{R_5R_6}$ remains approximately constant in a wide range of
858 variation of R_6 , producing a wide plateau range of varied R_6 , where a variation of $C_{R_5R_6}$ is very
859 small.

860 5. Conclusions

861 The theory of reduction of effective rotational Hamiltonians, developed by Watson [1–4, 23],
862 stands as one of the foundational cornerstones of high-resolution molecular spectroscopy, because
863 it connects basic molecular properties, such as geometry, masses and PES, to compact oper-
864 ator expressions, suitable for their precise fitting to experimental MW and high-resolution IR
865 spectroscopic data. This theory relies on the well-converged, power-ordered sequence of unitary
866 transformations $U_n = \exp(i\hat{S}_{2n+1})$, that gradually transforms a full theoretical rotational Hamilto-
867 nian of an asymmetric top, parameterized by the complete model sets of parameters in Cartesian
868 or cylindrical representation, to a reduced form with lower, experimentally determinable number
869 of constants (five quartic, seven sextic, *etc.*). Although Watson's straightforward theory (A -
870 reduction) is normally well conditioned for a typical semi-rigid molecule, cases of nearly-prolate or
871 oblate limits require an alteration of the scheme (S -reduction). However, for a rare case of quasi-
872 spherical tops, the denominators, defining the conventional reduction parameters may become
873 abnormally small, producing large transformation parameters, which results in poor convergence
874 of the perturbative expansion.

875 The first-order unitary transformation of the Hamiltonian, $H'(s_{111}) = \exp(i\hat{S}_3) \cdot H \cdot \exp(-i\hat{S}_3)$
876 is mathematically equivalent to an infinitesimal molecule-fixed J -dependent frame rotation. It
877 induces coordinated change in three rotational and six quartic constants $(B_\alpha, T_i) \rightarrow (B'_\alpha, T'_i)$ that
878 cannot be physically observable, while only five independent combinations of quartic constants
879 remain physically meaningful. In fact, in the first order the entire s_{111} -parameterized family of
880 Hamiltonians $H'(s_{111})$ produces the same rotational spectrum. Thus, the physically meaningful

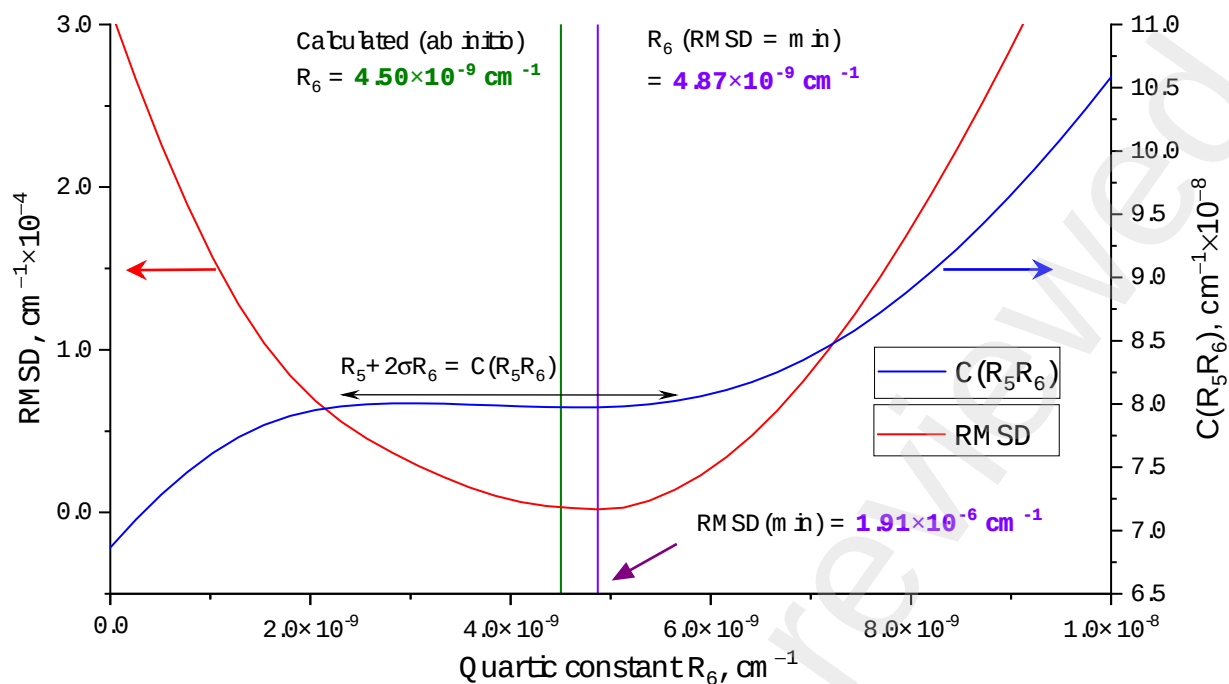


Figure 5: Fitting effective Hamiltonian $H_{\text{eff}}(R_6)$, consisting of six unreduced constants $D_J, D_{JK}, D_K, \delta_J, R_5, R_6$ and seven sextic S -reduced constants, using SO_2F_2 MW ground state dataset from Rotger *et al.* (2003) [17] with $J \leq 85, K_a \leq 50$. The value of R_6 was fixed at each plotted value and A, B, C , the remaining five quartic constants, and the seven sextic constants were refitted every time. The RMSD minimum is related to the truncated model and does not define an ‘observed’ value of R_6 .

object is not a particular decomposition $H' = H'_2 + H'_4$, but the entire family of constants, lying along the *gauge orbit* and controlled by s_{111} . Watson’s elimination of one quartic coefficient can therefore be interpreted as a gauge fixing: a particular choice of a representative within the equivalence class of spectroscopically indistinguishable (isospectral) Hamiltonians.

In this research work, we performed a critical analysis of a number of arguments, given in Sarka’s *et al.* (2000) [13] work, which supported the claim of a possibility to experimentally determine all six model quartic centrifugal distortion parameters of the quasi-spherical molecule SO_2F_2 . The system of arguments, provided there, purposefully defends the main claim. Our fundamental analysis unequivocally proves that although all six centrifugal distortion constants may indeed become numerically determinable in a framework with model constraints at sextic level, which fixes the first-order gauge parameters s_{111} , none of Sarka’s arguments prove the true physical significance of six ‘independent’ quartic centrifugal distortion constants.

There is an evident phenomenological similarity between misinterpretations, associated with the redundancy of six quartic centrifugal distortion constants and molecular harmonic force constants, expressed in redundant sets of internal valence coordinates [58–61] and it is worthwhile to recall the main point of the latter controversy. If R denotes a redundant set of molecular linear internal displacement coordinates and Q is its independent subset, then we can define a projector matrix $P = T(T^\top T)^{-1}T^\top$, where $T \equiv \partial R/\partial Q$. Armed with P , the fact of existence of infinite number of harmonic force constant matrices $\mathcal{F}_R$, describing true (‘observable’) physically identical force field, can be presented in the form $\mathcal{F}_R = \mathcal{F}_0 + L - PLP$, where $\mathcal{F}_0$ is any suitable starting force constants matrix and L is an arbitrary self-adjoint matrix [59]. Although this indeterminacy has several useful applications (e.g., ensures better transferability of individual model force constants), some works still claim that ‘this force field has physical significance’ [61]. A proper solution in

both cases would be making a definitive distinction between the true physical significance and interesting properties, appearing in model cases.

Another quite practical result is finding the first order invariant combination of quartic constants $R'_5 + 2\sigma R'_6 = C_{R_5 R_6}$, where primes denote the invariance after the reduction. In other words, introducing the constraint $R'_6 = 0$ in the effective Hamiltonian means that the fitted constant R'_5 can be interpreted as a linear combination of these model constants before reduction: $R'_5 = R_5 + 2\sigma R_6$. This fact provides a comprehensive clarification on the question of the ‘physical significance’ but in fact model character of simultaneous presence of R_5, R_6 in the effective Hamiltonian.

A practical and physically meaningful solution for quasi-symmetric tops like SO_2F_2 can be introducing the high-level theoretical anchors that remove the instability of fitting. *Ab initio* calculation of quartic and sextic unreduced centrifugal distortion constants produces values, corresponding to a chosen coordinate system representation and such constants correspond to $s_{111} = 0$. To ensure a well-conditioned LSQ fitting of SO_2F_2 , one must select five independent quartic parameters, which can be achieved practically in two ways:

(a) Calculate the six unreduced geometric quartic constants using a high-level harmonic force field. Strictly constrain one off-diagonal quartic term (e.g., R_6) to its *ab initio* value, and fit the remaining five. This safely locks the gauge without relying on truncation error curvature.

(b) As recognized in the framework of the Irreducible Tensor Operator (ITO) formalism [17], the ambiguity must be broken analytically by setting the symmetry-breaking parameter $t_9 = 0$. This defines the coordinate axes to be aligned with the principal axes of the distortion tensor.

Finally, we want to draw a special attention to possible general risks of introducing specific model constraints of fitted parameters with relying on small resulting confidence intervals for remaining fitted parameters. At least, in such cases it may be necessary to compare fitted parameters with their *ab initio* counterparts for proving physical significance of obtained sets of experimental parameters. The latter procedures have much wider scope of applications as a lack of experimental data is a universal problem and cutting of effective rotational Hamiltonians by fixing some low- and high-order constants is done very often.

6. Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used ChatGPT 5.6 Sol in a role of a handbook on mathematics and physics, as well as for searching most relevant literature sources. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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